International Association of Oil & Gas Producers

REPORT 2023sh | JUNE 2024

DATA SERIES

Safety performance indicators – 2023 data – High potential event reports

Acknowledgements

IOGP thanks those companies that have participated in the data collection programme.

This report was produced by the Safety Committee.

Feedback

IOGP welcomes feedback on our reports: publications@iogp.org

REPORT
2023sh
JUNE
2024

DATA SERIES

Safety performance indicators – 2023 data – High potential event reports

Revision history

VERSION	DATE	AMENDMENTS
1.0	June 2024	First release

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AFRICA ONSHORE

DATE: 04 Jan 2023
COUNTRY: Algeria
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Maintenance, inspection, testing
RULE: Energy isolation

NARRATIVE:

A high potential event occurred when unbolting a pressurized flange. During the assembly and disassembly of blind flanges of the cold box bypass, the GTP performing crew were mistaken about bypass flanges pass A 18" and worked on the blind flanges of pass B which were not already isolated and were under nitrogen pressure. When the crew observed a pressure release, they stopped working and left the area. Afterwards, the exploitation operator was in the area, recognized the situation, and quickly opened the vent to depressurize the line.

WHAT WENT WRONG:

- inadequate planning/risk assessment/preparation/supervision
- permit not specific without details
- toolbox talk content unknown and ineffective

Supervisor absence: job supervisor did not pause work to confirm the next flange to replace with the area operator.

Actual/potential consequences, causes, and actions:

So far, the causes identified are ineffective permit system and ineffective supervision:

- no task description available (e.g., job step description), tags were not listed
- permit with vague description approved, HSE revalidation was not marked
- toolbox talk was not documented, content of discussion unknown
- no toolbox talk after lunch break
- the supervisor chose to work on the flange under pressure without confirming with the area operator

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Perform a gap analysis on the system to improve the quality, communication, and compliance of the permit to work process.
- Create an improvement action plan that is endorsed by management.

Immediate recommendations for short term:

- Attend most permit to work (PTW) meetings to improve quality.
- PTW job description to be clear and work location identified.
- Breaking containment: operator to identify location and prove that the system does not contain pressure nor inventory (e.g., hydrocarbons, chemicals) by opening drains and vents, witnessed by the work team.
- Crew change: relevant incoming staff to countersign permits to ensure they are aware of the activities.
- Permits to be checked for:
  - quality of job description
  - job steps available
  - RA specific for the job

AFRICA ONSHORE

   - documented specific risks and controls on toolbox talk (TBT) for the job (or job steps if needed)
   - TBT to be held at start of work, after breaks, and when new job steps are added
   - TBT to be dated and signed by toolbox (TB) lead
- Late permits to be challenged and scheduled for later if not urgent*

** urgent: having an immediate impact on safety or immediate severe impact on quality or quantity of gas export*

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: **18 May 2023**
COUNTRY: **Egypt**
FUNCTION: Unspecified
CAUSE: Unspecified - Other
ACTIVITY: Transport - Land
RULE: Driving

NARRATIVE:

Driver observed to be unfit and falling asleep while driving. During the journey back to the office (from the base site), passengers were concerned that the driver was not concentrating on his driving and falling asleep at the wheel.

WHAT WENT WRONG:

A driver employed by a car rental company was observed as medically unfit for long trips. Consequently, they needed to be replaced with another driver who is in better medical condition.

During the journey, passengers were concerned that the driver was not concentrating on his driving and falling asleep at the wheel. However, no effort appears to have been made to stop the car and insist on a change of driver or report concerns to the Duty Manager.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Ensure drivers are fit to drive and are given adequate breaks during long journeys.
- Consider another form of transport such as flying/train.
- This is a serious near miss, giving a clear indication that using the car rental company for crew changes is high risk with a high potential of injuring personnel. The length of time endured on the road by the driver, in this case 10.5 hours, is unacceptable. It is therefore recommended that crew changes by the rental company are no longer permitted. All crew changes will be made by air and the short drive completed by road (the rental company). This will significantly reduce the risk.
- Encourage and promote our staff to speak up during unsafe working conditions.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Fatigue

AFRICA ONSHORE

DATE: **14 Jul 2023**
COUNTRY: **Nigeria**
FUNCTION: Drilling
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Drilling, workover, well operations
RULE: Line of fire

NARRATIVE:

The injured person (IP) was trying to lower the deck during which a moonpool A frame (150kg) was obstructing the path. While attempting to reposition the A-Frame with the chain hoist, the A-Frame fell on the IP, impacting the hard hat and wedging the IP between the A-Frame and pollution tank.

WHAT WENT WRONG:

- The IP was working alone while repositioning the A-Frame that was obstructing the downward lowering of the Texas deck.
- Non-compliance with approved procedures: stop work instruction from supervisor, a minimum of 2 people are required to reposition the A-Frame.
- Use of defective equipment (worn out chain hoist and missing safety latch) by IP to reposition the A-Frame.
- Improper position: IP was in the line of fire when the chain hoist used in the repositioning of the A-Frame snapped and impacted on IP's hard hat.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Enforce strict compliance with stop work instruction by site leadership.
- Need for frequent review of lifting accessories to ensure only fit for purpose are deployed.
- Awareness of line of fire risk.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

AFRICA ONSHORE

DATE: **29 Jul 2023**
COUNTRY: **Nigeria**
FUNCTION: Drilling
CAUSE: Falls from height
ACTIVITY: Construction, commissioning, decommissioning
RULE: Working at height

NARRATIVE:

A Hydraulic workover unit (HWU) that was rigged up on top of wellhead fell 49 feet to the ground with 3 workers in the work basket.

WHAT WENT WRONG:

The HWU tilted from its vertical axis and fell due to the following causes:

- The casing-wellhead connection strength was greater than the total installed weight on the casing.
- The casing head housing thread had a defect in its threading pre-installation.
- The defect in the casing head threading prevented the complete make-up of the casing head housing threads to the casing pin thread. The casing pin threading had a fracture after installation.
- The casing pin threading fracture reduced the yield strength of the casing-wellhead connection.
- The wellhead collar threading disengaged from the casing threading causing the unit to fall.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Need to identify any underlying issues in the casing and wellhead prior to activity.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: **08 Oct 2023**
COUNTRY: **Somalia**
FUNCTION: Construction
CAUSE: Unspecified - Other
ACTIVITY: Transport - Land
RULE: Driving

NARRATIVE:

At 4:42 pm, as a company rental vehicle was returning to the hotel after purchasing supplies for next day lunch boxes, the driver swerved to miss a child riding a bike. The vehicle left the road, hit a power pole, and rolled over twice before coming to rest on the side of the vehicle.

The vehicle's speed immediately prior to the incident was recorded at 106 km/h by the IVMS. The maximum speed limit is 60 km/h.

The driver and passenger (member of the public) were picked up by a company vehicle soon after the incident, travelling in the opposite direction and taken to the hospital.

Both the driver and his passenger were discharged from hospital 3 hours later with no significant injuries reported.

Company driver was on duty (but had been on standby) on the day of this incident which was due to low JMP activity.

AFRICA ONSHORE

WHAT WENT WRONG:

Excessive speed:

The primary immediate cause of the incident was the vehicle's speed, recorded at 106 km/h, significantly exceeding the maximum speed limit of 60 km/h. This reckless driving behaviour greatly increased the risk of loss of control and collision.

Failure to obtain permission:

The driver departed from the hotel premises without obtaining permission. This unauthorized departure deviated from established procedures and compromised the accountability and oversight necessary for safe operations.

Unauthorized passenger:

Another critical deviation was the driver's unauthorized act of allowing a third-party passenger into the company vehicle. This not only violated company policies but also introduced an additional distraction and potential safety hazard.

Non-compliance with seatbelt policy:

The failure to enforce the wearing of seatbelts by both the driver and the passenger significantly increased the risk of injury and compromised occupant safety.

Lack of adherence to security procedures:

The driver did not follow security procedures, particularly those outlined in the journey management plan (JMP). By deviating from established protocols, the driver compromised safety and security measures designed to mitigate risks.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Reinforcing the importance of strict compliance with company policies and Driving - Life Saving Rule.
- Ensure that all passengers are wearing seatbelts before journey starts.
- Check that vehicles are thoroughly inspected before rental and prior to departing on all journeys, and that all reported defects are addressed.
- Consistent monitoring of IVMS data and prompt intervention when deviations are detected.
- Produce in-vehicle rules stickers and SOP for drivers.
- Drivers are not allowed to pick up any un-authorized passengers from outside of the organization.
- Driver shall obtain clearance from Security department prior to departure and provide regular progress updates.
- Update journey management plans and LTMS regarding vehicle age limits.
- Implement driver check tests to be carried out throughout the project.
- After training in defensive driving, carry out refresher training after 6 months.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)

PEOPLE (ACTS): Following Procedures: Work or motion at improper speed

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

AFRICA OFFSHORE

DATE: 12 Feb 2023
COUNTRY: Angola
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Maintenance, inspection, testing
RULE: Line of fire

NARRATIVE:

The Rope Access Team was working in bay 8 inside the 3 centre cargo oil tank when they heard a loud noise in bay 9. The team stopped to investigate. They found an anode lying at the bottom of bay 9, near the central walkway, approximately 2 metres from bay 8. The anode had fallen from the top of the tank.

WHAT WENT WRONG:

- The anodes inside the 3 centre cargo oil tank were more than 20 years old, as it was installed on tanker age.
- There was no specific document or record of inspection done during the period of tanker or when it was converted as floating production storing and offloading (FPSO) in 2004.
- Most of the anodes in the upper and middle sections of the tank were found to be between 80% and 100% depleted.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Tank inspection procedure/statement of work (SoW) to consider requirement to perform intervention on obsolete anodes.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 19 Nov 2023
COUNTRY: Angola
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Production operations
RULE: Line of fire

NARRATIVE:

One section of a large pipe clamp was found lying on the deck of Module 40, between heating medium circulation pumps B and C. Further assessment confirmed that the clamp had fallen from the discharge line of heating medium circulation pump B. The clamp fell from an estimated 6 metres, and weighed 9.5kg.

WHAT WENT WRONG:

- Failure to secure, check, and monitor: two clamps of the heating medium pump B discharge line were not properly installed, and this led to their gradual loosening.
- Sudden temperature variations: the temperature of the pipeline from which the clamp section detached can vary from ambient to 120°C.
- Continuous vibration: the pipeline from which the clamp fell is located close to pieces of equipment which produce vibration, and this led to a gradual and unobvious loosening of the nuts and then the associated bolt.

AFRICA OFFSHORE

- Nuts not properly tightened: the nuts of the affected clamp and the second one installed only a few centimetres away were probably not tightened enough.
- Installation of the bolts: the orientation of the bolts allowed the clamp section to drop following the nuts detachment.
- Management systems failed to identify the risk (missing nuts) in time to prevent the incident.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Conduct full Dropped Objects Prevention Scheme (DROPS) sweeps on the topside modules, involving rope access.
- Consider modifying the DROPS inspection schedule so that it more exactly and strictly defines the requirement that meets industry standards. Communicate updated schedule to all affected personnel. Consider implementing audits or assessments to check that the inspection schedule is being followed.
- Conduct engineering study of the different types of clamp design/installation and recommend the most effective design/installation for the heating medium pumps lines.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: **11 Jul 2023**
COUNTRY: **Gabon**
FUNCTION: Drilling
CAUSE: Unspecified - Other
ACTIVITY: Maintenance, inspection, testing
RULE: Work authorization

NARRATIVE:

Half of a non-slip mat located around the rotary table pushed into the open rotary. The section of mat weighed 36kg and fell 22 metres to the mezzanine deck inside the wellbay.

WHAT WENT WRONG:

Unclean work area; control of work.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Hazard recognition, work authorization.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

AFRICA OFFSHORE

DATE: **06 Jul 2023**
COUNTRY: **Gabon**
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Lifting, crane, rigging, deck operations
RULE: Safe mechanical lifting

NARRATIVE:

Uncontrolled lifting operation during vessel backload.

WHAT WENT WRONG:

Improper operation by crane operator.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Retraining of crane operator.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

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DATE: **18 Apr 2023**
COUNTRY: **Ghana**
FUNCTION: Production
CAUSE: Cut, puncture, scrape
ACTIVITY: Maintenance, inspection, testing
RULE: Line of fire

NARRATIVE:

Whilst disconnecting a section of hydraulic tubing from a valve actuator, the tubing was inadvertently released under pressure and came into contact with the IP, who was in the line of fire. IP sustained laceration to the right cheek and nose.

The job was stopped, and the IP was escorted to the onboard medic. After onboard medical assessment, the IP was sent onshore for further treatment (Medevac).

WHAT WENT WRONG:

The changing out of the O-ring on the leaking joint on the SDV was not well planned. A permit was not in place to cover the entire O-ring changeout scope, which would have included tubing disconnection. This resulted in the failure to properly assess the overall scope and ensure all risks for the scope were identified and mitigated. The job was only communicated to a few directly involved project team members, and not to relevant stakeholders, such as the Area Authority or Production Superintendent. As a result, the required Control of Work (CoW) authorities were not aware of the scope and therefore unable to validate that appropriate risk controls were in place. This failure is underscored by the lack of adequate coordination between the projects and operations team, which would have fostered an enhanced team approach to project execution.

IP was only focused on giving assistance, and failed to enquire on the task ongoing. This resulted in him poorly judging the situation by only concentrating on catching potential oil drips into the bucket he decided to assist in holding, resulting in him placing himself in the line of fire.

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CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Conduct lessons learned sessions covering all crew members, highlighting the potential consequences of misuse of the PTW system, based on this incident.
- Review contractor personnel selection and requirements, assess relevant gaps, and develop a plan to improve the process.
- Conduct refreshers/awareness on Line of Fire, Energy Isolation, and Work Authorization (Life-Saving Rules).
- Conduct competency validation for all assigned isolation authorities on the FPSO competency register to confirm adequacy of meeting relevant expectations.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 15 Oct 2023
COUNTRY: Ghana
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Maintenance, inspection, testing
RULE: Unspecified

NARRATIVE:

While a scaffold supervisor was inspecting/assessing hard barriers at upper levels, a corroded guardrail was seen lying on the grating deck at Upper level 2, beneath CM/SW Exchanger-C cooling medium outlet line and seawater inlet line. Upon further assessment, it was noticed that a section of guardrail on the mezzanine deck at level 2 had broken off from its position due to corrosion at the welded support base. The nature of the drop that could have led to its unusual positioning on the grating beneath the pipes on the lower deck was undetermined. The weight of the guardrail was estimated to be 22kg and the mezzanine deck was approximately 3.4 metres from 5P upper level 2. There were no injuries or environmental harm as a result of this incident.

WHAT WENT WRONG:

- Handrail condition at the time of incident: Externally, there was visible paint chipping because of the potential impact of the handrail with other structures within the location from drop. Coating of an adjacent flange in the location of the drop was observed to be broken. Some amount of dent was observed on the handrail, albeit minimal.
- Cause and type of corrosion identified: after the incident, inspection of the failed base of the handrail was conducted which revealed extensive corrosion at the welded base. Historical leak of seawater from the GRE line onto the handrail structures which was situated directly below it.
- Continuous vibration was noted in the area. Vibration measurement revealed a reading of 18mm/s in that module, which is likely to affect the already corroded handrail.

AFRICA OFFSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Conduct a survey and identify handrails on topsides, with single point of attachments that are subjected to excessive vibration, LSW, and corrosion.
- Secure/brace standalone sections of handrails with single points of attachment at the base to other structures or clamp to adjacent handrails.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: **30 Mar 2023**
COUNTRY: **Ghana**
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Production operations
RULE: Line of fire

NARRATIVE:

On an offshore supply ship, in the field, a reel traction drive system was being prepared, ready to move the system astern to connect to the next umbilical reel (static umbilical SEUMB6). After preparing some hydraulic hoses and ensuring no snag points on the trackway, the operators left the system. About 15 minutes later, a noise was heard and subsequently two pieces of handrail were found on the main deck (weighing 0.9kg and 4kg) that had fallen approximately 8.5 metres from the top of the traction reel system.

WHAT WENT WRONG:

The combination of poor design and subsequent loading of the handrail, led to the failure and dropped object.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Processes developed to assess all non-fixed equipment that are brought onboard, to ensure that they are fit for purpose in terms of safety when working at height and that they are suitably designed to eliminate loading of any lifting gear.
- When redesigning the platforms and their associated lifting arrangement, there's a need to ensure that they do not clash with ladders and other parts of the structure and/or introduce additional hazards like pinch points.
- Once a design is approved, the procedures will require to be updated to offer clear guidance on storage when not in use and not left to circumstance or opinion.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

AFRICA OFFSHORE

DATE: **30 May 2023**
COUNTRY: **Nigeria**
FUNCTION: Production
CAUSE: Falls from height
ACTIVITY: Production operations
RULE: Working at height

NARRATIVE:

A tanker approached the pilot boarding area and a fast crew boat arrived to perform the tanker clearance. The tanker had a combination ladder arrangement consisting of a pilot ladder and an accommodation ladder. The terminal pilot along with 3 other people successfully climbed up the pilot ladder before it was the turn of the IP (a terminal agent). A few steps before reaching the accommodation ladder, the IP was feeling exhausted and uncomfortable and started descending back to the fast crew boat. On his way down, he lost his grip of the pilot ladder and fell into the sea with the crew boat being clear from his fall. His life jacket inflated on contact with the sea and he was rescued.

WHAT WENT WRONG:

The terminal agent (IP) let go of the pilot ladder while descending.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Reduce exposure by restricting pilot ladder personnel transfer to tasks which require physical presence onboard export tanker. Review justification and investigate remote alternative solutions as far as possible.
- Personnel boarding a ship should be in good physical condition and feel confident for climbing the pilot ladder and stepping on the accommodation ladder.
- Where possible, consider a health check by medical personnel for personnel boarding vessels.
- Develop and complete training for the pilot and crew of the crew boat on checking that the pilot ladder and the accommodation ladder is arranged correctly.
- The vessel should make a lee to facilitate boarding. The vessel should be stopped without making way, or sail at a heading and speed agreed between the vessel Master and the pilot of the crew boat to make a good lee.
- Use STOP card when conditions are unsafe or if any defect in boarding arrangements is reported and not corrected.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Fatigue

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

AFRICA OFFSHORE

DATE: **15 Mar 2023**
COUNTRY: **Senegal**
FUNCTION: Drilling
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Drilling, workover, well operations
RULE: Line of fire

NARRATIVE:

Operation was pulling 26" BHA on 5-7/8" drill pipe through water column. While break out procedures were commencing, the hydraulic power tong operator's attention was drawn to a closed-circuit television (CCTV) screen, ensuring the pipe racker was aligned to secure stand to be racked. The power tong was moving forward during this time and trapped the IP against the stump. The IP was tying the slip handles at the time. When attention returned to power tong operations, the operator immediately retracted the power tong, suspending operations.

WHAT WENT WRONG:

- Redzone expectations potentially articulated unclearly at corporate level.
- Change to process assessment may have been inadequate.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Floor hand failed to identify he was in the line of fire.
- Contractor moved hydra-tong without confirming no personnel in line of fire.
- No formal risk assessment was carried out to ensure hazards were highlighted and addressed before changing operating method.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Organizational: Inadequate communication

ASIA/AUSTRALASIA ONSHORE

DATE: 10 Jul 2023
COUNTRY: Australia
FUNCTION: Drilling
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Drilling, workover, well operations
RULE: Line of fire

NARRATIVE:

While running in hole with 3 ½" tubing, two Floor hands (FH) pulled the slips out of the rotary table master bushing and placed them on top of the table, towards the rear of the derrick. One FH left the area to prepare the next joint of tubing. At this time, while the tubing was being run into the hole, the other FH crouched down in the exclusion zone between the descending tubing and the rear of the derrick. The FH was attempting to reposition the slips on their own then lost control, causing the slips to drop back into the master bushing, inadvertently setting the slips, which caused the tubing to rapidly bend towards the FH. The Driller initiated an emergency stop of the top drive. There were no injuries to personnel.

WHAT WENT WRONG:

- Personnel working within rig floor exclusion zone.
- Work party supervisor failed to enforce operations exclusion zone.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Supervisors must ensure that all site personnel are aware of exclusion zones within their work area and strictly enforce exclusion zone rules.
- Ensure exclusion zones are included in risk assessments, understood, and complied with by all personnel.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

ASIA/AUSTRALASIA ONSHORE

DATE: **08 Mar 2023**
COUNTRY: **Australia**
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Production operations
RULE: Line of fire

NARRATIVE:

Shortly after a new fin fan pulley was installed, it dropped to grade, making superficial contact with two workers in close proximity. No injuries were sustained.

The pulley dropped onto scaffolding approximately 1.6 metres below and weighed 86kg.

WHAT WENT WRONG:

The pulley was not properly secured to the fan shaft. An Allen key was used to secure the grub screws instead of a torque wrench.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Complete a risk assessment and develop a maintenance procedure relating to the installation and replacement of fans and motor pulleys for cooler fans.
- Review permit issuing and monitoring criteria for shutdown activities to ensure that work packs and associated procedures/methodology are adequate.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

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DATE: **06 Dec 2023**
COUNTRY: **Australia**
FUNCTION: Production
CAUSE: Struck by (not dropped object)
ACTIVITY: Lifting, crane, rigging, deck operations
RULE: Line of fire

NARRATIVE:

Whilst a work crew was moving a scaffold frame with a telehandler, the rigger/scaffolder was walking near the load, stabilizing it. The rigger lost his footing and fell, and whilst he was getting up, his foot was struck by the front wheel of a telehandler, causing a crush injury.

WHAT WENT WRONG:

- Personnel involved in the incident did not adequately assess the risk of collision between the telehandler and rigger/scaffolder.
- The work crew did not implement an adequate exclusion zone between the rigger/scaffolder and the mobile plant.

ASIA/AUSTRALASIA ONSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Review and upgrade mobile plant risk assessments to address hazards and controls associated with pedestrians and mobile plant.
- Review and update procedural controls associated with exclusion zones and spotter responsibilities.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 25 May 2023
COUNTRY: Australia
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Production operations
RULE: Unspecified

NARRATIVE:

Internal combustion occurred inside the air assist system of the Stage 1 Flare Warm Wet.

WHAT WENT WRONG:

- Design review: The risk of hydrocarbon ingress into the air assist system was not adequately considered during the Hazard and operability study (HAZOP), after the change in the flare system operating strategy.
- Corrective action – needs improvement: Risk of hydrocarbon ingress into a flare air assist annulus was identified in a previous event with lessons potentially not clearly articulated.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Standards: before beginning a task, familiarize yourself with primary and secondary muster points.
- Risk identification: vendor engagement during front-end design and safety reviews/HAZOP specifically for novel or complex systems is important.
- Communication: operating procedures to provide key vendor parameters and a clear line of sight to operational risks.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

ASIA/AUSTRALASIA ONSHORE

DATE: 13 Jan 2023
COUNTRY: Australia
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Production operations
RULE: Unspecified

NARRATIVE:

A beam pump polished rod parted during normal operations.

The failure of the polished rod caused three objects, parted polished rod section (19.4kg), rod rotator (18.1kg), and load cell (3.5kg) to be released and fall to the ground, adjacent to the wellhead.

WHAT WENT WRONG:

- The shaft had failed due to tensile-tensile or unidirectional bending fatigue failure in the condition of high stress with high stress concentration.
- Bainitic microstructures were evident throughout the thickness of the rod, indicating possible inadequate heat treatment.
- Significant banding was found in the core regions of the rod.
- A significant difference was found between the current results and the hardness data provided in the heat certificate.
- The hardness results shown in the heat certificate were also inconsistent with the expected ultimate tensile strength of the steel.
- The impact strength of the steel was very low for a grade 4140 steel.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Undertake an As Low As Reasonably Practicable (ALARP) evaluation to determine if existing controls are sufficient. Identify if additional controls are required to be at ALARP.
- Risk Management: ensure additional controls (drop zones/2-minute time limit) are effectively communicated and applied.
- Inspection/audit: verify that controls are known, implemented, and effective in the field.
- Independent review/inspection by a third-party inspection (failure analysis).
- Organizational failure to adequately learn: similar incidents have occurred within the business, however, after a review of the incidents and associated actions it was identified that they were only partially effective, not fully implemented, and/or not known.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

ASIA/AUSTRALASIA ONSHORE

DATE: 19 Dec 2023
COUNTRY: Papua New Guinea
FUNCTION: Drilling
CAUSE: Pressure release
ACTIVITY: Drilling, workover, well operations
RULE: Bypassing safety controls

NARRATIVE:

When backloading nonaqueous drill fluid (NAF) from the rig mud tanks to tanker trailer, the tanker became over pressurized causing the rear hatch to release, detach from the body of the tank, and come to rest 13 metres away in front of the truck. As per procedures, two workers were standing on the rear left side of the tanker at the time of the incident and sustained minor first aid injuries because of the pressure release. Approximately 48bbls of NAF were released from the trailer.

WHAT WENT WRONG:

- Tank was not protected by a pressure relief valve.
- Tank hatch was not vented in accordance with procedure for transferring fluid using external pump.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Re-certify pressure relief valve and re-install on tanker trailers.
- Develop separate job aids for transfers with external pumps and tanker internal pumps highlighting required venting configuration.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

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DATE: 13 Nov 2023
COUNTRY: Thailand
FUNCTION: Drilling
CAUSE: Falls from height
ACTIVITY: Drilling, workover, well operations
RULE: Working at height

NARRATIVE:

Operation was in the process of run completion (tubing 2-7/8") to 1164 mAH of 2004 mAH. While the driller lowered the traveling block holding the 2-7/8" tubing, the IP was working in the rig floor area, attempting to hold the tubing with his hand. IP was carrying the tubing, and attempting to hold the tubing for removing the thread protector. The unexpected movement of tubing caused the IP to fall from the rig floor to the catwalk. The rig floor was 3.6 metres from ground level. The catwalk was 0.8 metres from ground level. The IP fell 2.8 metres.

WHAT WENT WRONG:

From the incident simulation, the YT elevator handle caused the bending of tubing (stored energy) and then released the energy to the IP, which caused him to fall through the V-Door.

ASIA/AUSTRALASIA ONSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Design: The V-shape stopper used to support and lock the travelling block set during rig down for mobilization caused a potential obstruction. YT elevator is normally used due to the more robust locking mechanism. However, the handle of the YT elevator stuck out, creating a risk of obstruction by the V-shaped stopper. Due to operation, the safety chain needs to be removed for the purpose of tubing pick-up activity. Verify the impact of the V-shaped stopper on the mast structure. Remove the V-shaped stopper. A simulation should be run to see if the risk of obstruction has been reduced. Issue a Management of Change (MOC) for the redesign of the latching point for traveling block mobilization. Consider to study the appropriate elevator which is suitable for the flushby unit working condition. Consider alternative fall prevention, e.g., a sliding door or something designed to minimize the opening gap and ensure no one can fall from height and issue an MOC for modification.
- Procedure and safe work practice: The current practice is to handle pipes by hand. Rope or other means for pipe handling would minimize the risk of line of fire and hand on load hazard. Share practice and review work instructions among units under Well Services.
- Human factor: Time out for safety was not effectively implemented to verify the abnormality causing more vibration. No one on the rig floor stopped IP from performing the rig floorman task. Communicate the importance of stop work authority (SWA) and taking time out for safety to all operations.
- Supervision: Toolpusher/Driller did not assign roustabout to replace floorman, causing an unauthorized person to perform a task. There is no clear role and responsibility for interfacing the scope between the rig floor and the completion team. Set up ground rules for manpower arrangements during break time and communicate to all crews. Communicate role and responsibility to all completion and workover crews.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Disabled or removed guards, warning systems or safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 25 Oct 2023
COUNTRY: Thailand
FUNCTION: Construction
CAUSE: Exposure electrical
ACTIVITY: Construction, commissioning, decommissioning
RULE: Work authorization

NARRATIVE:

There was work on the installation of a beam pump. The worker wanted to cut CB07 cable that was supposed to be spare. The mistake was that he cut CB10 (live line). This cable had electricity running through it and was supplying power to one working beam pump. This incident led to a short circuit that cause a flash and explosive sound. No one was injured.

WHAT WENT WRONG:

The excavation was not entirely completed and the sand was not all cleared; some parts of the cables were obscured by sand, leading to cutting the wrong cable.

ASIA/AUSTRALASIA ONSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- There was no written work instruction to define the authorized person for cable cutting. Cable continuity/voltage check requirements were not in the work instruction. Develop/revise the electrical cable-cutting procedure to include work steps, tools, safety precautions, authorized persons, and cable identification.
- Unclear electrical work limitation for incompetent person. Identify the list of electrical works and determine the required competencies, authorization, and supervision for works.
- The foreman did not check the work permit details when a new task was assigned after permit issuance and made an assumption when picking the cable to cut. The right cable should be identified by digging to the cable end.
- No cable identification was marked on the conduit. Need to mark the cable on the conduit (sleeves) externally and make this a requirement for a new project.
- The work permit should have been revised by the electrical engineer after the addition of cable cutting to the scope.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

ASIA/AUSTRALASIA OFFSHORE

DATE: 13 Jan 2023
COUNTRY: Australia
FUNCTION: Production
CAUSE: Exposure electrical
ACTIVITY: Maintenance, inspection, testing
RULE: Energy isolation

NARRATIVE:

De-ballast pump was found not electrically isolated, following removal and reinstatement of suction/discharge spooling for flange face inspections. The connecting rod from the isolator handle to the isolation switch was not fully engaged.

During the preceding 13 days, flange face inspections/repairs continued, with seven different permit holders, without any further verification of the isolation that was in place.

When the Isolating Authority applied the isolation and turned the isolation handle to the 'locked open' position, as per the approved isolation scheme, the internal connecting rod was not fully engaged and consequently slipped, leaving the isolation switch in the 'closed' position. Isolation was independently verified as per requirements, which included cross-checking of the electrical isolation certificate with the appropriate equipment. Verification does not involve, nor require, proving isolation integrity.

WHAT WENT WRONG:

- Isolation was ineffective for the equipment, as the isolation handle did not engage fully with the connecting rod and subsequently the isolator switch.
- The isolation standard and associated manuals and procedures are not stringent enough in enforcing the requirements for 'Proving an Isolation'.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Clearly define requirements in the isolation manual around proving isolation integrity and provide examples, including the requirement to 'try test' before starting work. This should take place with the work party and the area authority on initial (first issuance) permit sign-on. This also demonstrates to the work party that the equipment is unavailable and provides an added layer of assurance.
- Effective immediately, during the normal isolation process for LV isolators with connecting rods, prior to the application of lock and tags, open motor control centre module and confirm connecting rod has operated correctly with isolator in the 'OPEN' (Isolated) state. This will also provide another level of assurance for that particular isolation, as the isolator switch/device can be visually inspected to confirm it is in the 'OPEN' (isolated) state.
- Investigate options for potential modifications to the interconnecting rod and handle latch mechanism to increase mechanism reliability.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

ASIA/AUSTRALASIA OFFSHORE

DATE: 06 Feb 2023
COUNTRY: China
FUNCTION: Production
CAUSE: Struck by (not dropped object)
ACTIVITY: Production operations
RULE: Other issue – no applicable rule

NARRATIVE:

During the berthing of a workboat at the drilling platform, the vessel was unable to effectively control its position and collided with the platform.

WHAT WENT WRONG:

Direct Causes: The vessel was unable to effectively control its position and drifted, resulting in a collision with the drilling platform.

Indirect causes:

- inadequate risk analysis
- inadequate emergency response measures
- inadequate implementation of regulations

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Notify the accidents, improve operational planning, minimize the number of vessel berthings, and enhance awareness of protecting against berthing operation risks.
- Organize specialized discussions on the accident, strengthen the ability to identify risks, and handle emergencies during upwind and nighttime berthing operations.
- Organize training on the implementation of regulations and requirements to enhance awareness and capabilities in implementing system regulations on site and in the use of equipment.
- Improve the operation manual, specify the restricted conditions for nighttime operations with poor visibility, upwind and upstream operations, and strong wind conditions.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

ASIA/AUSTRALASIA OFFSHORE

DATE: **14 May 2023**
COUNTRY: **China**
FUNCTION: Production
CAUSE: Water related, drowning
ACTIVITY: Diving (incl. decompression), subsea, ROV
RULE: Bypassing safety controls

NARRATIVE:

During the berthing of a platform, a vessel was improperly operated by the captain. During high-speed sailing, the captain operated the left and right rudder paddle handles to turn 180 degrees and reverse thrust and deceleration, the two main engines stopped successively, and the ship lost control and collided with the platform. It caused damage to two sea pipe sections of the platform through the deck and two pile legs, as well as damage to the windows and guardrail of prospecting ship's steering room.

WHAT WENT WRONG:

The deputy captain of the exploration ship failed to implement the relevant provisions of the "Operation Code for Ships berthing offshore Facilities" and the "Safety Management Rules for Guard Ships" regarding ships entering the 500 metre range of offshore facilities when performing the task of docking platform. Improper operation led to the overloading of the main engine and the loss of control of the ship, resulting in the ship colliding with the platform.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- It is recommended to publicize safety warnings for ships under charter in the East China Sea, strictly implement the requirements of the system, check the integrity of key equipment, and ensure the safety of ship operation.
- It is recommended to further strictly implement the provisions for ships to enter the 500 metre safety zone of offshore facilities, and further strengthen the training, supervision, and inspection of ship-related personnel.
- It is suggested to comprehensively sort out the ships with full turning rudder angle that are currently leased, conduct special training for key crew members, and only after passing the test can they be boarded. At the same time, the functional transformation of large rudder angle limit under high load operation of the ship type is completed as soon as possible.
- It is recommended to post a safety warning "It is forbidden to conduct large rudder angle steering operation when the ship is sailing at high speed" in the obvious position of the bridge for the rudder propeller full rotation ship. It is suggested to further strengthen the record of key crew members and increase the qualification examination of service ship types. For captains who do not have the operation experience of full rotary ship types with rudder OARS, they need to have qualified captain guidance, and can operate the ship independently after the evaluation of the guiding captain and the ship company.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Following Procedures: Work or motion at improper speed

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective Personal Protective Equipment

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PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 04 May 2023
COUNTRY: Indonesia
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Production operations
RULE: Unspecified

NARRATIVE:

A foreign tanker crossed within one nautical mile of a normally unmanned platform, and the radio operator did not immediately notice the radar alarm. On becoming aware of the situation, the radio operator immediately contacted the tanker, which changed course away from the platform.

WHAT WENT WRONG:

The radio operator did not immediately notice the alarm. The investigation revealed that the alarm was not loud enough, and the radio operator was distracted as he was preparing the day's POB.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Importance of 24/7 radio room coverage with competent people.
- Implementation of a formal handover process.
- Issues with inadequate alarm volume.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

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DATE: 05 Sep 2023
COUNTRY: Philippines
FUNCTION: Production
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Maintenance, inspection, testing
RULE: Energy isolation

NARRATIVE:

During a maintenance activity, lube oil leaked from a back-up pump's discharge line when said pump, that is part of a generator package, ran while the package was isolated. Said pump ran upon disconnection of an instrument cable. The back-up pump was missed as part of the overall electrical isolation of the generator package.

WHAT WENT WRONG:

Back-up pump was missed out as part of the overall electrical isolation of the generator package.

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CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Ensure that all isolation points are considered when isolating a package. This includes looking at all relevant drawings and assessing all items included in a package's system.

CAUSAL FACTORS:

No Causal Factors allocated

DATE: **25 Jul 2023**
COUNTRY: **Philippines**
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Production operations
RULE: Unspecified

NARRATIVE:

As the facility was experiencing high winds, the signage of the facility located at its highest level was damaged and a portion of the signage dropped onto a lower level. The object was enough to cause a fatality, as per the drops calculator.

WHAT WENT WRONG:

Corrosion on the signage together with high winds.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Assess all hanging objects for possibility of falling due to high winds and the possibility of them causing fatalities if they drop.

DATE: **30 Jan 2023**
COUNTRY: **Thailand**
FUNCTION: Drilling
CAUSE: Falls from height
ACTIVITY: Drilling, workover, well operations
RULE: Working at height

NARRATIVE:

IP proceeded to bow deck to cross to the platform by using the Personal Access Ramp (PAR) to service equipment at Mud Logging unit. At that time, two deck crews were standing by at the area to assist IP to go up to the platform by adjusting the portable staircase to proper position. First the IP used his left hand to hold onto the PAR and stepped his right foot up onto the PAR. Then IP moved his right hand to grab the handrail on PAR staircase while the tender was heaving downward (making the PAR move upward relative to tender), causing the IP to not be able to position himself properly on PAR stair, lose his grip on the PAR, and fall approximately 2 metres to the bow deck. The crews who were Standing by in the area immediately approached the IP, assisted him clear of the area and escorted him to the rig clinic for further assessment.

WHAT WENT WRONG:

IP moved his right hand to grab the handrail on the PAR staircase while the tender was heaving downward (making the PAR move upward relative to the tender), causing the IP to not be able to position himself properly on the PAR stair, lose his grip on the PAR, and fall approximately 2 metres to the bow deck.

ASIA/AUSTRALASIA OFFSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- PAR procedure does not clearly identify the proper level (PAR relative to tender) before crossing up and down. Amend PDR for safe use of PAR to include safe passage of PAR. Specific height limitation and add up roles and responsibilities of watchmen to communicate safe PAR usage to all passengers when at AMBER level.
- Safe PAR briefing/induction does not clearly identify the detail of 3-points contact technique to climb up and step down. Install the Safe PAR Usage Signage Board, at BOW deck. Develop a safety PAR induction VDO and share to tender rigs to implement and use in the induction.
- Inadequate hazard communication. Revise safe PAR briefing package at induction and communicate to all crews and confirm competency of safe PAR usage in SSE checklist.
- Human Factor: Perceived as Norms. Emphasize correct size of PPE and STOP work authority to request for PAR lower down when unsafe for all personnel onboard.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: 10 Sep 2023
COUNTRY: Thailand
FUNCTION: Construction
CAUSE: Falls from height
ACTIVITY: Construction, commissioning, decommissioning
RULE: Unspecified

NARRATIVE:

While recovering the unused anchor chain, the weather picked up with squall making the vessel stand by on weather with some part of the anchor chain on board and the rest hung down from the stern to seabed. The AB was on deck to secure the fire hose that fell off the box by the waves. The 2.5m-3.5m swell came from the stern, flooded the main deck starboard, causing the vessel to list to starboard, and pushed the AB overboard.

WHAT WENT WRONG:

During the operation, there was a squall passing the GBN location with max. wind 37-40 knots, with sea and swell height of 2.5m-3.5m. The large swell hit the stern causing huge flooding on the whole main deck and swept the AB overboard.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Lack of deck watch duty: enhance deck watch duty compliance.
- Lack of project planning (missing of marine expert advisory): include the marine expert in the project development process.
- Lack of project risk assessment involved with vessel suitability vs. weather limitation: conduct risk assessment including vessel suitability review vs. adverse weather.
- Lack of emergency disconnection plan: improve the vessel emergency disconnection plan.

ASIA/AUSTRALASIA OFFSHORE

CAUSAL FACTORS:

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 28 Nov 2023
COUNTRY: Thailand
FUNCTION: Construction
CAUSE: Unspecified - Other
ACTIVITY: Construction, commissioning, decommissioning
RULE: Unspecified

NARRATIVE:

The tower operator noticed a smoke detector alarm activated on the fire alarm control panel. He promptly checked and discovered that the room was filled with smoke. The operator immediately informed the HTL barge foreman and Safety Officer, and a rigger responded to the location, successfully extinguishing the fire originating from the wall electrical outlet.

WHAT WENT WRONG:

Rainwater from the window dripped to the wall electrical outlet and caused a short circuit.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- No risk assessment for this hazard of the barge condition – porthole leak: Develop/revise the concern risk assessment to include the hazard of porthole leak and risk of water ingress to wall outlet socket, set up the control measure, and communicate the risk assessment and control measure to all relevant contractor staff.
- Inappropriate inspection method for the porthole: There is no procedure for porthole leak inspection; only a random room visit is done during hygiene inspection which does not mention porthole inspection. Develop the procedure (or safe work practices) and programme for periodic porthole inspection.
- The chargers for handheld radios were left connected and unattended in the accommodation room: Have a dedicated location for radio charger in the area that will not charge the radio unattended.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

EUROPE ONSHORE

DATE: **05 Feb 2023**
COUNTRY: **Austria**
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well operations
RULE: Line of fire

NARRATIVE:

An object fell from the mast onto the rig floor. A hoop, which had been mounted to protect the lamps, fell from the mast and onto the rig floor, while braking down and laying down 5 1/2" drill pipes.

No personnel were hit by the falling object.

WHAT WENT WRONG:

- insufficient fixing of hoop
- deficient welding joint
- vibration and abrasion on hoop

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- replan fixing of the hoop on the rig mast
- relocate mount of rope
- extend dropped object inspection on fixings

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: **20 Oct 2023**
COUNTRY: **Austria**
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well operations
RULE: Other issue – no applicable rule

NARRATIVE:

Sudden shot of downhole camera head during dismantling: A camera run with slickline was carried out on the gas probe. The camera was disassembled in the workshop of the slickline trailer. When the camera head (diameter approx. 40mm, height approx. 30mm) was loosened, the head shot off and damaged the windscreen of a company vehicle parked next to the trailer.

WHAT WENT WRONG:

- Hazard identification: Failure to identify the hazard of trapped pressure in the camera housing.
- Operations and maintenance: Absence of detailed work instructions including hazards and precautions, absence of maintenance and inspection routines for the camera housing.

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- Design: Absence of reliable method to release possible trapped pressure from the camera housing. Missing engineering calculations supporting if the design of the camera housing is suitable for the intended purpose.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Update and amend JSAs for downhole camera activities in view of trapped pressures. Review JSAs of other slickline/wireline activities where the hazard of trapped pressure may occur.
- Develop detailed work instructions for downhole camera activities, including description of hazards and precautions.
- Develop and implement written maintenance and inspection routines for downhole camera equipment items.
- Verify the design of the camera assembly in view of future downhole camera operations.
- Investigate a safe and reliable method of releasing potential trapped pressure in the camera housing before opening the camera assembly.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 23 Jan 2023
COUNTRY: Germany
FUNCTION: Production
CAUSE: Struck by (not dropped object)
ACTIVITY: Maintenance, inspection, testing
RULE: Line of fire

NARRATIVE:

Employees of a workover contractor were at a well site to collect equipment used in the moving of a workover rig. The workover activities had been completed and the pumpjack had been put back into operation. The contractor's grounding cables still had to be disconnected from the grounding point at the base of the support structure within the enclosed area of the pumpjack. The injured employee, a short service employee, entered the area which was fenced off with an unlocked gate to remove the grounding cables. While doing so, his working clothes were caught by the moving counterweight of the crank arm and he was squeezed between the support structure and the counterweight, sustaining injuries to his head and upper body.

WHAT WENT WRONG:

- No machine risk assessment had been performed for the pumpjack to identify and implement effective technical barriers.
- The risks from simultaneous operations had not been effectively assessed.
- No risk-based onboarding process and supervision programme for new employees had been implemented by the contractor and no formal minimum requirements for qualification and experience for new employees had been established.

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CORRECTIVE ACTIONS AND RECOMMENDATIONS:

The event emphasized the importance of adhering to risk assessment processes related to the safe operation of equipment and to all work activities, including an assessment of simultaneous operations. It also emphasized the importance of specifying minimum competency and onboarding requirements for new employees and of verifying the effective implementation prior to work beginning.

Recommendation:

Perform machine safety assessments for all machinery and implement sound risk assessment processes for all types of work occurring onsite. Implement control measures resulting from the risk assessments.

Build minimum contractor requirements for workforce qualification requirements and onboarding processes and ensure contractors are following requirements.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 03 Apr 2023
COUNTRY: Norway
FUNCTION: Construction
CAUSE: Dropped objects
ACTIVITY: Construction, commissioning, decommissioning
RULE: Line of fire

NARRATIVE:

A grave lock coupler that was used to fasten a tarpaulin on TP deck had loosened and fell down. Wood was used between steel and coupler to protect the paint on the steel. This led to friction between the wood and the coupler, which resulted in it loosening over time. Coupler fell to site near an elevator. The area below was not cordoned off as there were no activities for work at height. No injury to personnel or material damage.

WHAT WENT WRONG:

- Wood was used between coupler and steel to protect the paint.
- Couplers were not checked if done correctly after being placed.
- Couplers had been mounted on steel frame for a long time (unsure for how long).
- No written routines for checking couplers used to fasten tarpaulins.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Ensure all risks related to working at height are mitigated.
- Ensure that all equipment being used is in good technical condition and that it is used correctly.
- Ensure that actions are being taken to secure loose objects and report potential dropped objects.

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CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 15 Mar 2023
COUNTRY: Norway
FUNCTION: Construction
CAUSE: Dropped objects
ACTIVITY: Construction, commissioning, decommissioning
RULE: Line of fire

NARRATIVE:

Surface operators were going to work inside a tank (bottom of tank). Work team had to vacuum up sand inside the tank. The hose to be used was already in place from the previous shift. When vacuuming, the work team discovered that the hose was clogged. Work team exited the tank to unclog the hose that was placed at the top of the tank (TU deck). Electricians then went inside the tank.

Hose was shaken to remove its content. In doing so, the hose hit a shovel that was placed inside the physical barrier. The shovel fell through the tank opening. The electricians witnessed the fall and impact. There were personnel near where the shovel landed in the tank.

WHAT WENT WRONG:

- Equipment and tools at the top of the tank were not secured.
- Team did not establish a barrier at the bottom of the tank.
- Poor housekeeping attitude/judgement by storing loose items near opening.
- Implemented routines were not followed (SSS, Take 5, risk assessment regarding securing objects at height).
- The area around the hole was not checked before shaking the hose.
- Supervisor and Field Supervisor on the shift the incident occurred had not signed the Safe Job Analysis prior to work.
- Neither Operators nor Area technician checked the workplace after end of work and before start of work.
- Operators did not sign out of the tank when they went outside to shake the hose (entry list).

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- More follow-up by the line organization and consequential response for poor attitudes.
- More follow-up and guidance from Supervisors in the field.
- Provide more and better training to Area Technicians. Have a better understanding on what is defined/included as ''workplace''.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

EUROPE ONSHORE

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 26 Oct 2023
COUNTRY: Norway
FUNCTION: Construction
CAUSE: Struck by (not dropped object)
ACTIVITY: Construction, commissioning, decommissioning
RULE: Line of fire

NARRATIVE:

During a site tour at a pipeline production facility, a near miss occurred when a company representative was standing within an observation area/walkway, close to the bevelling station. The area in question had a physical barrier (gate), which had been zip-tied in an open position.

Access into the area was part of the tour, and production was ongoing. A solid steel fence was in place but did not function as a barrier due to the pipe ends protruding over and past the fence. The pipe was moving in automated sequence, laying stationary for most of the time. Having observed the pipe bevelling through a plastic strip curtain, the company representative was engaged in a conversation when a pipe was moved from the bevelling rack to the mainline holding rack. The pipe was moved to a position such that the end was exactly where the person had been standing 10 seconds earlier. He was facing the opposite direction, so did not see the pipe until it passed closely by his right arm. The final spacing between pipes was approximately 15cm.

WHAT WENT WRONG:

- lack of risk evaluation of the situation/multiple personnel within a restricted area
- lack of control of accompanying personnel
- insufficient barrier not separating product from personnel
- lack of attention, personnel placing themselves in line of fire

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Warning sign placed on the entry gate and the zip-tie removed.
- Adjustable cordoning erected to restrict access to product.
- If safety barriers are circumvented, ensure mitigating measures are implemented before proceeding.
- Do not approach machines or product without explicit permission from host.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

EUROPE ONSHORE

DATE: 20 Jan 2023
COUNTRY: Romania
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well operations
RULE: Other issue – no applicable rule

NARRATIVE:

During drilling activity at 84m deep, due to high surface vibration, the back lock clamp from an elevator (200gr) dropped on the rig floor from approximately 2 metres. No personnel were in line of fire and nothing was lost in the hole.

WHAT WENT WRONG:

- use of worn-out lifting/handling devices
- welding joint in vibration environment, usually not recommended; mechanical failure
- possible internal structure defects

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Revision of handling/lifting equipment.
- Supplier manufacturer to provide an improvement proposal for back lock clamp, by design and fabrication.
- Search for local supplier manufacturer as back-up supplier.
- Request procedures for handling equipment from provider.
- Ensure that the drilling contractors have a procedure provided by the original equipment manufacturer (OEM) which includes the inspection, maintenance, and quality control (QC) of the elevator.
- Risk awareness mentioned in JSA for handling/lifting.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 20 Apr 2023
COUNTRY: Romania
FUNCTION: Drilling
CAUSE: Unspecified - Other
ACTIVITY: Drilling, workover, well operations
RULE: Bypassing safety controls

NARRATIVE:

During N2 lifting operation with coiled tubing, the connector flange between injector head and BOP started to leak releasing gas, N2, water, and mud into the air. The site crew reacted and secured the well. Well shut-in under monitoring, the pressure was 104 bar.

EUROPE ONSHORE

WHAT WENT WRONG:

- Audit procedure for Drilling, Workover and Well Service Units was not followed.
- Sand control was not required based on production history.
- High drawdown created by N2 and increased chock size on manifold.
- Lack of knowledgeable leadership on well site of company personnel - Controlul manifestarilor eruptive was not followed.
- Poor risk awareness: Well control training/certification of company personnel missing.
- Role and responsibility insufficiently defined: JOD of supervisor is not specified relating to well control procedures.
- Poor safety culture on site.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Increase awareness and leadership of supervisors.
- Specify and communicate maximum allowed drawdown and information about target formation for future jobs orders for planned N2 lifting/unloading operations with contractor.
- Perform sand control risk assessment for Sarmatian Basal formation for particular field before started well completion equipment.
- Offer well control training and certification for supervisors and chief operator of crew.
- Review responsibilities of supervisor to be more specific relating to mandatory competency and responsibilities.
- Implement motivational management on both companies; consider consequences management for repeated deviation from HSSE basic requirements (like PPE).
- Implement safety culture improvement programme

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

PEOPLE (ACTS): Use of Protective Methods: Personal Protective Equipment not used or used improperly

PEOPLE (ACTS): Use of Protective Methods: Disabled or removed guards, warning systems or safety devices

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

EUROPE OFFSHORE

DATE: 12 Apr 2023
COUNTRY: Denmark
FUNCTION: Production
CAUSE: Water related, drowning
ACTIVITY: Diving (incl. decompression), subsea, ROV
RULE: Work authorization

NARRATIVE:

A Diving Support Vessel (DSV) was engaged in saturation diving operations adjacent to a platform. Two divers were engaged in moving a transponder subsea for the purpose of metrology for a new spool to a pipeline from the Subsea Safety Valve Structure.

The DSV's taut wire system, consisting of metal wire with a clump weight, had already been deployed from the DSV to provide a vertical reference for the DSV dynamic positioning system. The two divers were pulled off the seabed (-42m water depth) due to recovery of the starboard taut wire. The divers ended up at -12 m water depth before the taut wire clump weight recovery was stopped. The divers immediately freed themselves and returned to the diving bell and the dive was terminated. The divers were brought back on the DSV. The divers were medically examined, and no ill effects were identified. Preventive treatment was administered.

WHAT WENT WRONG:

Immediate Causes:

- The potential for umbilical entanglement with starboard taut wire was not identified as a hazard by the Dynamic Positioning Officer (DPO) at point of executing the taut wire retrieval operation.
- Open comms system was in place but muted and set to 'closed circuit' – delaying the bridge team's reaction to the lifting of the divers.

Analysis:

- Decision to retrieve starboard taut wire was made to remove hazard for divers during the dive.
- DPO not aware of divers' entanglement.
- The bridge team made the incorrect assumption that the Dive Supervisor was aware of the taut wire recovery – it was recovered without final repeat back and approval by the Dive Supervisor.
- It had become customary practice to recover taut wires when divers were 'seen' as being in a 'Position of Safety' without verbal confirmation from the divers.
- There was a gap in the management system- taut wire 'weight' lifting operations, which were not covered by a lift plan.

Root Causes: Management System Inadequate - as a result, taut wire operation was not treated as high-risk operation:

- There is no operational procedure/checklist/lift plan for taut wire operations.
- Taut wire was identified as a hazard (DP Operations Manual), however no controls were in place to mitigate the hazard.
- Not International Marine Contractors Association (IMCA) compliant for lifting operations whilst diving operations ongoing.

EUROPE OFFSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Review of management system to identify gaps and verification of the system onboard DSV during operation should be done.
- Lift plans for taut wire deployment/recovery to include step to "confirm divers are in a place of safety" before start of operations should be done.

Actions to be done to avoid communication issues:

- Ensure communication system is functioning correctly between DSV bridge and Diving Control Room prior to start of activities.
- Ensure communication is in clear and concise language to avoid any potential for misinterpretation.
- Ensure commands and/or instructions are repeated back prior to carrying them out, to verify understanding.
- If communication is not clear or there are any doubts on the command issued, an ALL STOP should be called.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: 16 May 2023
COUNTRY: Germany
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Maintenance, inspection, testing
RULE: Work authorization

NARRATIVE:

A cable tray cover of 5kg was not reinstalled and left on the gangway grating. A storm picked the cover up and the cover fell 4m onto a gangway beneath.

WHAT WENT WRONG:

- Due to various cable pulling works, cable tray covers are regularly removed and reinstalled. During this assembly and disassembly, one of the covers was not reinstalled or secured in a safe manner.
- Exposure to adverse weather conditions.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

No follow-up check by company supervisor to ensure that the covers were all installed correctly. Employee did not reinstall or store the cable tray cover safely at the end of his shift.

Recommendations:

- Define safe storage place for cable tray covers.
- Ensure a safe workplace at the end of shift or when leaving (end of work).
- Install secondary retention line on covers to avoid dropped objects.
- Employees receive instructions for cable tray cover handling.

EUROPE OFFSHORE

- Item added to job safety analysis.
- Supervisor shift handover documentation assurance and regular site visits.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured
PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 27 Dec 2023
COUNTRY: Netherlands
FUNCTION: Drilling
CAUSE: Unspecified - Other
ACTIVITY: Drilling, workover, well operations
RULE: Safe mechanical lifting

NARRATIVE:

The drill crew onboard the Well Safe Protector were in the process of pulling out of the hole after successfully cutting the 9 5/8" casing.

As the crew had reached the final stand, which was the cutting assembly, a lifting sub was installed and the stand was lifted to approximately 4 metres above the rotary, where the cutting blades of the tool were inspected.

Once the inspection was completed, the "link tilt" was functioned to lay out the assembly, which caused the cutting assembly to swing towards the v-door, and, due to a mismatched thread connection with the lifting sub and box connection of the cutting assembly, the assembly, weighing 300kg, came loose and fell approximately 1.14 metres from its lowest point (bull nose) and then toppled forward.

The Night Tool Pusher (NTP) and Neptune DSV immediately stopped the operation to investigate the incident, where it was discovered that the lift sub remained in the BX elevator, positioned about 4.7 metres above the drill floor.

As the crew was investigating the incident, the elevators where then inadvertently opened, causing the lifting sub, weighing 37.7kg, to fall approximately 4.7 metres.

There were no injuries to personnel and a full investigation was initiated.

WHAT WENT WRONG:

- Inadequate communication: Numerous situations and opportunities were observed/heard by the involved and present team operating within the rig floor environment. These indicators were not acted upon when the situation warranted. Communications were not accurately achieved or verified, and this led to assumptions being made about activity stages being achieved correctly, which was not the case.
- Poor visibility: Conditions of the windows during the investigation confirmed contamination presence (residues) and therefore a reduction in visual clarity to both the Driller and A/D chairs, particularly to the Elevators which had implications for this event.
- Failure of stopping the operation when initial issue with making up lifting sub: Numerous witness statements refer to an awareness of issues being experienced during making up of the equipment by the Floor Hand, however, no visual confirmation was achieved, and verbal responses received to non-specific questions portrayed the correct equipment was being fitted and merely a thread issue was the initial problem.

EUROPE OFFSHORE

- Risk Perception Poor: Following a review of the witness statements and the latter investigation teams review of the event, it became apparent that a significant number of opportunities/indicators for corrective intervention (STOP THE JOB) were presented prior to the incident occurring, however, these were either not recognized as such and/or acted upon, including those of present Service Company and Client personnel throughout various stages of the task. Example of this was when the Floorhand was attempting to make up the incorrect lift sub, it was noticed by a few personnel and was not stopped and checked fully. Had this been done perhaps they would have recognized this was not suitable.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Planned TOFs built into the procedure and an additional TBT would have had better planning. Therefore, the plan would have been discussed and equipment looked out prior to getting to the point of laying out the Cutting Assy.

Windows on Doghouse if required to be cleaned during operations then stop, clean the windows before continuing. There were also cameras available at the time which were not switched at the time from Monkeyboard level to just above rig floor level.

When something does not look right then we must stop the task and question what's wrong, not incorrectly assume someone else should know and just go with one individual's comment, multiple supervision at the worksite at time and all missed this opportunity. Similar to action point one, better planning and communication would have picked up what correct equipment was required.

Communications if impaired at any stage, then pull the crews into the Doghouse where information can be clearer and explained correctly. Call a TOF prior to continuing operations.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: **03 Oct 2023**
COUNTRY: **Norway**
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well operations
RULE: Line of fire

NARRATIVE:

A person got stuck in the derrick elevator, as it stopped going down due to parted power cable. The person donned a harness and managed on his own to escape through hatch through the roof of the elevator.

The elevator has a failsafe function with both primary and secondary breaks. The elevator is not held up by the power cables but runs on tooth racks and sprockets running up and down the derrick. The risk of the elevator cabin falling further is not deemed as a potential consequence.

Three substantial cable guide brackets, each weighing 5.65kg, for guiding the power cable, dropped down to the deck below. One dropped 8 metres and the other two fell 4.8 and 5.9 metres, respectively. The brackets could potentially have fallen to deck levels further below, resulting in kinetic energies in the order of 2800 to 3400 Joules. The deck levels below were not barriered off.

EUROPE OFFSHORE

WHAT WENT WRONG:

The cable trolley derailed during a period of high winds causing the signal cable to the elevator being ripped off. The brackets for holding the signal and power cable in place were pulled by the force of the signal cable causing the cable brackets to drop down and the elevator to shut down.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Review design weaknesses and ensure more robust design (e.g., fastening of brackets/secondary retention) of the cable guides/brackets before installing further brackets to prevent dropped objects.
- Consider installing a "de-railing" kit.
- Establish a process for quality assurance of maintenance done to ensure the maintenance is in accordance with requirements.
- Consider re-visiting the wind criteria for use of the elevator.
- Look at further improvements to facilitate better shielding of the elevator's external environment (wind, etc.).
- Consider implementing wind sensor to stop use of elevator in high winds.
- Ensure experience transfer towards rigs with same type of derrick elevator and make sure that technical improvements to the elevators are captured.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

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DATE: 07 Dec 2023
COUNTRY: Norway
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
RULE: Safe mechanical lifting

NARRATIVE:

It was discovered that a removable railing section that is an integral part of the helideck baggage handling lift arrangement was missing. The railing section was located at the main deck level, presumably after having been caught by the wind (gusts up to 40 knots and helicopter downdraft). The weight of the railing section is 6.2kg and it dropped 25m to the main deck level. It is not known when the incident occurred, but no personnel were in the area at the time of the incident.

WHAT WENT WRONG:

- The railing section had not been properly fastened inside the cargo lift arrangement.
- Lack of training with regard to the importance of properly installing the railing inside the lift arrangement.

EUROPE OFFSHORE

- Strong winds caused the railing section to fall.
- Deficiency in the design of the cargo lift.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- The railing section has been secured by installation of a safety wire.
- A windwall has been installed in the lift arrangement.
- A review will be carried out to look at potential further improvements in design and lifting procedures.
- Ensure the personnel is properly trained in using the cargo lift before commencing such an operation.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PEOPLE (ACTS): Use of Protective Methods: Disabled or removed guards, warning systems or safety devices

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 22 Jun 2023
COUNTRY: Norway
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
RULE: Line of fire

NARRATIVE:

A scaffolding component fell down in connection with a blind lift operation. After the cargo had been released from the hook and the signaller had given the order to lift, the crane hook got caught on a scaffolding spire. Scaffolding spires with vange and cantilevers came loose and fell as one component. Personnel inside the cordoned-off area saw the object come loose and ran for cover. The signaller sought cover behind some waste containers, but the scaffolding component hit a pipe arrangement on the way down, changed direction and landed on the waste containers. The signaller was hit on the shoulder by the scaffolding component as it tipped over. The actual consequence of the incident is assessed as personal injury - medical treatment injury. It is considered that under slightly different circumstances, persons within the cordoned-off area could have been struck by the scaffolding component in a different way with fatal consequences. The incident is classified with the highest severity possible, fatality.

EUROPE OFFSHORE

WHAT WENT WRONG:

Direct causes:

- Hook caught on the scaffolding component after "free-hook" signal. - Signaller's unfortunate escape route.
- Mechanic inside barrier, under the crane hook in motion.

Underlying causes:

- Risk of crane hook getting caught wasn't identified.
- Placement of cargo not carried out in accordance with original plan.
- Inattention in connection with "free-hook" signal.
- Inadequate planning of lifting operation involving blind lifts.
- Lift considered as routine lift. New obstacles not considered.
- Scaffolding work permits not managed by area responsible.
- Lack of knowledge of and compliance with procedures and requirements.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Key learning for other units - Mainly operational causes:

- Verify requirements, guidelines, and aids.
- Strive to achieve greater compliance with requirements and guidelines through training and verification.
- Develop a practical training programme for all involved in lifting operations with an offshore crane, incl. the operational lifting manager.
- Develop confirmation activities in accordance with the practical training programme.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

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DATE: 22 Apr 2023
COUNTRY: Norway
FUNCTION: Production
CAUSE: Pressure release
ACTIVITY: Production operations
RULE: Work authorization

NARRATIVE:

Prior to the installation of the new hook-up spool, a blind lid on a new flowline to well was to be split and removed. Confined pressure was not detected, and pressure was therefore not drained before the split was performed, resulting in lifting the blind lid (34 kg) 1.6m up in the air and hitting the operator in the face on the return. 2.4kg HC gas escaped, resulting in gas detection, 2 sensors activated. This led to activation of the deluge system and ESD. One person in the work team was seriously injured.

WHAT WENT WRONG:

Immediate causes:

- Splitting was carried out with pressure in the flowline.
- The work permit was activated without the correct plan for isolation (ICC) for the work being prepared, linked to the work permit and put in place (live).

EUROPE OFFSHORE

- Those involved incorrectly perceived that it had been verified that there was no pressure in the flowline.

Most important underlying causes:

- Drifting everyday practice in processing work permits, irrelevant ICC linked to work permit, and information in operational overview (Shift Vision) that was not taken into consideration.
- Verification of depressurized system at one verification point, and with incorrect verification method.
- The mechanics loosened the clamps in accordance with requirements, without getting any impact on the gas detectors, and therefore perceived that the system was depressurized.
- High operational tempo, quality in Work Order plans, and resource coordination between several interfaces resulted in strain on capacity for the operations discipline.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Ensure and verify integrity of the management of isolation certificates (ICC), through training and verification of methods for testing of depressurization and HC-detection, and correct ICC list linked to Work Permit - secure operational capacity for planned operations - mentoring/on-the-job training to increase operational competence - strengthen compliance of work requirements through self-assessment and peer review/verification.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 15 Aug 2023
COUNTRY: UK
FUNCTION: Production
CAUSE: Exposure noise, chemical, biological, vibration, extreme temperature
ACTIVITY: Maintenance, inspection, testing
RULE: Energy isolation

NARRATIVE:

A technician felt a release of pressure when changing a bushing assembly on a vessel. The vessel was isolated as part of an ongoing TAR and had been purged with nitrogen. This release was from a nitrogen enriched atmosphere and filled the technician's breathing space. The technician felt unwell and was monitored by the site medic.

The technician was using an air fed form of respiratory protection which takes in local air from waist height.

WHAT WENT WRONG:

The break of containment (BOC) checklist was followed but was not sufficient to prompt consideration of venting the isolation envelope to atmosphere to prove zero energy.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Update the BOC checklist written within control of work procedure CoW to include clarification on when to consider, prior to BOC, venting the isolation envelope to atmosphere to prove zero energy. Include within the 2024 review of CoW competency assessments, the learnings from this incident relating to how to prove zero energy prior to BOC.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

EUROPE OFFSHORE

DATE: **13 Jan 2023**
COUNTRY: **UK**
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Production operations
RULE: Unspecified

NARRATIVE:

A crew member discovered a section of cable tray cover lying on the deck, and its origin could not be immediately determined. It was later determined that the cover had fallen from the top module. The cable tray cover weighed 1kg and fell 16m. It had the potential to cause a fatality if it had hit someone.

WHAT WENT WRONG:

The retention clamps holding the cover in place were only suitable for indoor use. It is believed that regular contact between a crane wire rope and adjacent cable tray covers may have loosened cover retention fixings.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Improved hazard identification plus inspection and maintenance required.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: **06 Aug 2023**
COUNTRY: **UK**
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Production operations
RULE: Unspecified

NARRATIVE:

An N2 leak test was being conducted when a loud pop was heard from the deck above. Upon investigation it was found that the helium injection hose had burst completely. Nobody was injured, however there was the potential for one of the operators to be seriously injured as he was within close proximity. It was reported that hose restraints were in use, however due to the failure mode, they were unable to restrain the hose.

WHAT WENT WRONG:

Hose destruct testing indicates that the helium manifold valve was in the closed position resulting in rapid over-pressurization – operator error. Additionally, the helium pump was rated to 15 thousand PSI whereas the hose was only rated to 10 thousand PSI.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Test procedure and associated checklists should include improved requirements for external pressure test equipment/plant HP/LP interface protection management. Hose restraint requirements should be assessed and reviewed following failure of the hose restraint in this incident.

EUROPE OFFSHORE

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 21 Apr 2023
COUNTRY: UK
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Production operations
RULE: Other issue – no applicable rule

NARRATIVE:

During area tech morning walk around/checks, a smell of gas was reported at the platform underdeck area.

Gas was observed to be exiting the amine sump tank 8" overflow/vent line which is routed over the side of the platform. A gas monitor was used to measure gas concentrations on the P underdeck, and 3% LEL was recorded at a 5m distance from the overboard dump/vent line.

Following further discussions and investigation works, the source of the gas release was identified to be gas in the rich amine flash drum passing to the amine sump tank via the passing drain valve.

Following discovery of the release, operations then closed the DB&B in the upper leg (gas phase) of level bridle after which gas was no longer observed or detected at the amine sump tank vent.

WHAT WENT WRONG:

Original design did not take in to account the hazard of pressurized sight glasses being isolated from atmospheric tanks by only the use of a valve.

Individual failed to recognize that isolating the liquid phase on a sight glass would allow a route from the gas phase directly to the drains system.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Offshore Actions:

- Apply formal isolation to applicable valves to prevent recurrence of gas release whilst there is a known passing valve in the system.
- Undertake review of similar atmospheric tanks and apply formal valve isolations where required.

Onshore Actions:

- Identify all locations where a similar interface exists between process equipment and tanks with an atmospheric vent and raise an eMOC to determine a suitable engineering solution to mitigate gas migration risk (e.g., blanking drain lines or removing sight glasses if not required).
- Discuss event with peers to identify if they require a similar eMOC to be raised.
- Create lessons learned pack for sharing.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

MIDDLE EAST ONSHORE

DATE: **26 Feb 2023**
COUNTRY: **Iraq**
FUNCTION: Production
CAUSE: Struck by (not dropped object)
ACTIVITY: Maintenance, inspection, testing
RULE: Line of fire

NARRATIVE:

Head and arm injuries due to mobile platform collapse : a company electrical technician was conducting a routine check inside a compressor station switchgear room. A mobile scaffold was present close to one of the switching panels. The electrical technician attempted to move the platform away from the panel.

WHAT WENT WRONG:

As the scaffold was moved it fell backwards onto the technician, who tried to stop it falling on him. The IP was unable to hold the scaffold and it fell, trapping him on the ground. The injured technician managed to lift the scaffold off himself and leave the switchgear room. He called for help and was assisted by nearby people.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Greater awareness is required on appropriate actions for Emergency Response: The need to call the EROC number instead of the nearest clinic. There was confusion as to whether the ambulance was actually coming or not.
- Effective communication of hazards between work parties before work starts is a continuous improvement item. Effective communication with all parties could have prompted discussion on the placement of the mobile scaffold platform and the hazards associated with moving mobile scaffolding.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

DATE: **29 May 2023**
COUNTRY: **Iraq**
FUNCTION: Production
CAUSE: Struck by (not dropped object)
ACTIVITY: Maintenance, inspection, testing
RULE: Line of fire

NARRATIVE:

Individual struck by forklift just after 3am, during scaffold removal as part of the turnaround activities, a forklift (operated by the Company) collided with one of the contractor scaffolders.

The IP was transferring scaffolding materials from V-01 to an area approx. 4m away accessible by the forklift.

WHAT WENT WRONG:

The forklift ran over the IP’s foot leading to a subsequent fall and harm to the IP’s hand and potential crushed foot.

The IP was treated on scene by the medical team before being transferred to hospital.

Following review and treatment, the IP shows no broken bones, but a dislocated finger which was reset.

MIDDLE EAST ONSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Engagement with the workforce to understand the risks/hazards/potentials associated with vehicle movement.
- Greater supervision of the worksite to assist flagman in ensuring that the worksite is prepared for vehicles and all personnel are safely away from Line of Fire.
- Greater understanding of PTW and intent required.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Inattention/Lack of Awareness: Fatigue

DATE: 19 Oct 2023
COUNTRY: Iraq
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Construction, commissioning, decommissioning
RULE: Work authorization

NARRATIVE:

Arc Flash at substation: At around 14:20, contractor workers inadvertently opened the rear panel of the 3.3KV 12.3 system, mistakenly believing it to be breaker of 12.1-PM-09A.

The scope of work was for testing on a motor cable as part of 12.1 rehibition.

WHAT WENT WRONG:

The panel mistakenly selected was live operating at Breaker 12.3-PM-09A. As result, an immediate tripping of the circuit breaker of 33kV HS1.1 (12.1-PM-09A) outgoing feeder TMM1.2A occurred. No one was injured.

The access for the panel, containing the required cable connections, is via the rear of the breaker cabinets and there was poor lighting.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Clear Communication and Documentation
- Comprehensive Training on Electrical Safety
- Competent Personnel in Charge
- Continuous Monitoring and Supervision
- Regular Safety Audits and Inspections

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

MIDDLE EAST ONSHORE

DATE: **10 Apr 2023**
COUNTRY: **Kurdistan Region Of Iraq**
FUNCTION: Drilling
CAUSE: Falls from height
ACTIVITY: Drilling, workover, well operations
RULE: Unspecified

NARRATIVE:

While running the Upper Completion into the hole, the personnel on the rig floor heard something fall. Rig floor operations were suspended immediately to investigate the cause of the sound.

The HPU (Hydraulic Power Unit) located at the rear of the drill floor was observed to be missing one of its access doors. The right-hand door had been blown off the HPU due to a strong wind and had fallen from the rig floor to come to rest on the ground near the back side stairs of the rig floor. The door had impacted the stairs, and slid down the stairs, to land at ground level at the bottom. The weight of the door is 43kg and fell 9m to the ground.

Nobody was injured and no damage caused to any other equipment because of this incident.

The operation was stopped immediately, and the crews conducted a safety stand down – proceeded to check on all the doors on the rig floor to ensure all were properly secured, checked security of the handrails on stairs, and checked for any other loose objects on drill floor. The door landed near the High Pressure (HP) Kill Line, so the entire Kill Line was inspected. No HP Kill Line equipment was damaged.

A rigorous inspection of the mast, monkey board, rig floor, and sub-structure was also conducted to assess and identify any other potential issues or risks. Dropped Object Inspection has been initiated to make sure a suitable level of quality was applied.

WHAT WENT WRONG:

- There was no maintenance or checks conducted on the doors or their hinges or locking controls.
- No secondary retention on the doors. The door itself weighs 43kg and should have had a minimum of two secondary retentions installed.
- With such adverse weather conditions including strong winds the doors should have been checked so that they are secure during such conditions.
- In general, a lack of scheduled maintenance and bad choice of positioning of the latch to its current position and not considering if this will indeed secure both doors.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Where possible, hinged doors are to be replaced by sliding doors.
- The introduction of door restrictors on all opening-outward doors. Door restrictors would limit the opening distance thus eliminating any hazard of the door falling if it became detached.
- A suitable door closer to be fitted to all similar doors that would aid in self-closing.
- Implement Weekly inspections of these doors and also add all the doors (including monkey board escape door) to the Dropped Object inspection register. Preventative maintenance checks and registers are to be implemented and kept up to date.
- Correctly manufactured door hinges with locking pins to prevent lifting are to be fitted using a correct welding or securing method.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

MIDDLE EAST ONSHORE

DATE: 21 Aug 2023
COUNTRY: Kurdistan Region Of Iraq
FUNCTION: Production
CAUSE: Exposure noise, chemical, biological, vibration, extreme temperature
ACTIVITY: Construction, commissioning, decommissioning
RULE: Bypassing safety controls

NARRATIVE:

A scaffolder, working on the Amine Contact tower, was rendered unconscious by a suspected discrete flow of H2S. He regained consciousness when emergency life support apparatus (ELSA) was applied and then assisted to ground level.

WHAT WENT WRONG:

- Technical design incorrect.
- Monitoring of construction not effective - O2 sensor not installed in the original build.
- Risk assessment not effective - flue gas combustion profile not adequately understood.
- TRA did not identify the H2S emissions as a hazard.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Update TRA to include comprehensive controls for working over 6m in the LPG plant.
- Determine air/waste ratio settings once establishing oxygen monitoring in-line system.
- Determine correct chimney height for the chimney stack and acid vent stack.
- Use an industrial hygienist to undertake air monitoring to map potential exposures to flue gas emissions, and provide additional control recommendations.
- Create a comprehensive working at height education package detailing the hazards of CO2, H2S, and other emissions when working at height.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

MIDDLE EAST ONSHORE

DATE: 02 Jul 2023
COUNTRY: Kurdistan Region Of Iraq
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Excavation, trenching, ground disturbance
RULE: Work authorization

NARRATIVE:

Contractor excavator bucket made contact with the 24" export (sales gas) pipeline while trenching for the perimeter fence project. The result was a surface scratch to the pipeline coating.

WHAT WENT WRONG:

- Specifications or criteria incorrect - lack of as-built drawings and permanent demarcation of underground services.
- Lack of line detection equipment - resulting in underground service locations being incorrectly marked.
- Assessment of required skills or competency ineffective - reliance on general knowledge of personnel to location of underground services.
- Incorrect behaviour reinforced - practices ran counter to the TRA and ground disturbance procedures permitted to save time.
- Lessons learned not embedded - lessons learned from numerous line strikes not adequately captured within the control of work process.
- Leadership and accountability ineffective - permitted deviance from expectations in TRA.
- Risk analysis or tolerance not effective - hazard regarding buried hydrocarbon lines not adequately considered.
- Specified controls not followed - hand digging to first uncover and then mechanical excavation requirements not followed.
- Enforcement of SPP not effective - personnel performing the work not involved in the control of work process.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Strict adherence to the ground disturbance procedure process, to identify underground services and potential significant hazards before any ground disturbance/excavation takes place.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

MIDDLE EAST ONSHORE

DATE: 31 Aug 2023
COUNTRY: Kurdistan Region Of Iraq
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Production operations
RULE: Unspecified

NARRATIVE:

At 10:44:11 am a high H_2S alarm activated along with a blue strobe and horn indicating the presence of H_2S, followed by a high and high-high alarm from 3 other detectors at different times.

The control room operator informed the process supervisor of the alarm and HMI readings. The process supervisor instructed the operators to mask up and investigate the location of activated detectors. The CCR operator announced all other personnel to evacuate the plant area and make their way to the assembly point.

The masked-up operators detected a liquid DGA leak (visually) in the Amine System area and confirmed that the active alarms matched the leak location as shown on the HMI in the CCR. The leak was confirmed to be at the inlet line flange of the Rich Amine Flash Tank. The rich DGA is a two-phase stream at the inlet flange conditions of the tank.

At 10:49:55 am the Facility safety system (two fixed H_2S detectors) (five minutes from the initial detection) functioned effectively and caused an emergency shutdown (ESD) of the plant as per the Plant SD Key.

The operators responded under mask to the flange leak by closing the isolation valves U/S and D/S and closing the downstream valve of the flash tank, leading to isolation of the source of leak.

The DGA loss was estimated at 100 litres. This DGA led to contamination of the surrounding area which was estimated at 3 metres radius. In addition, the leak resulted in H_2S release to atmosphere.

Once the leaking section was depressurized and isolated, the Maintenance team assessed the inlet flange of the Flash Tank and found that the isolating gasket had deteriorated and caused the leak.

The Maintenance team prepared a permit to work and associated safety documents to replace the damaged gasket. They replaced all the Viton isolation gaskets installed in the Amine System with Flexitallic carbon steel standard gaskets as a precautionary measure. Once the gaskets had been replaced and the area was free of any acid gasses, the Amine unit was restarted.

WHAT WENT WRONG:

The investigation team found the MOC process/procedure is in place and includes a stage gate process which includes a comprehensive multidisciplinary review encompassing all the disciplines. The investigation team did not find any comments from the assigned reviewers for the MOC - Amine piping replacement.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- To ensure a thorough and comprehensive second-stage review for the MOC, by ensuring all the disciplines are included in the distribution. In addition, the progress to the next stage is not permitted until sign off from all disciplines is received. It's important not to limit the review period of MOC's to a specific number of days (14 days), however, sufficient time to be allowed for all assigned reviewers to provide their comments before the MOC proceeds to the next stage since some of the assigned reviewers may be on rotational leave when the MOC is submitted for review and comment.

MIDDLE EAST ONSHORE

- It is imperative to establish and consistently maintain all necessary documentation, including data sheets, piping specifications, isolation kits specifications, and any other relevant documents for our facilities and services and updating them regularly based on future projects. This practice is essential to prevent/mitigate incidents/accidents and avoid potential confusion.
- Procure/source materials from subcontractors who have the competency to choose the right materials for the given service.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: **14 Apr 2023**
COUNTRY: **Kurdistan Region Of Iraq**
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Production operations
RULE: Unspecified

NARRATIVE:

Following a process upset which commenced the previous day, there was a loss of containment (LOC) from a floating head gasket on a flange on the De-ethanizer Re-boiler. Early investigation efforts, which initially focused on the LOC, revealed that the release rate was 9,614kg/hr of condensate (predominantly in vapour phase). Further investigation revealed that during efforts to manage hydrates in the Cryo systems, De-ethanizer Column was subjected to pressure above the Maximum Allowable Working Pressure (MAWP) of 550psig, and outwith the maximum range of the pressure reading instrumentation (567psig), remaining in this state for a period of approximately 9.5 hours.

WHAT WENT WRONG:

- Lack of adequate overpressure protection on De-ethanizer column.
- Pressure Alarm High had been disabled in the SCADA system.
- No formal procedure/process in place currently for the management of hydrates withing the cryogenic system.
- High dew points when the Molecular Sieve bed exceeded adsorption time by 4 hours.
- Prolonged operation at high flow rates and high dew point.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Engineered measures required to prevent overpressure of the expander, de-ethanizer system to be implemented.
- Pressure transmitter to be used to provide an input for an executive action that isolates vulnerable sections from upstream pressure sources.
- Reinstate Pressure Alarm High at the earliest opportunity and develop an approved documented alarm response.
- Current methods used to 'warm' the system after the introduction of hydrates to be ceased with immediate effect.
- Review critical process parameters and how they might be better monitored by control room personnel to ensure the plant is maintained within the safe operating envelope.

MIDDLE EAST ONSHORE

- Review reinstatement of dew point trip to establish if this can be achieved with sufficient reliability of the instrumentation.
- Develop formal procedure for mol sieve operation including normal, abnormal, and emergency scenarios.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 03 Jul 2023
COUNTRY: Kurdistan Region Of Iraq
FUNCTION: Construction
CAUSE: Unspecified - Other
ACTIVITY: Construction, commissioning, decommissioning
RULE: Line of fire

NARRATIVE:

The erection of a steel compressor shelter had begun on the morning shift, although the structure was incomplete at lunch time with two sections (out of four) attached to the concrete foundations. A strong gust of wind blew through the site and the structural steel sheered from the foundations and fell on to two adjacent vessels. No persons were in the area at the time as lunch was being taken.

WHAT WENT WRONG:

- The beam's large surface area contributed to the force exerted by the strong wind.
- Structure inadequately secured.
- Inadequate supervision.
- Failure to follow method statement.
- Failure of the securing bolts.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Improve planning - the planning phase to consider the available space for installation and placement of equipment.
- Task supervision - supervision of task by the Performing Authority/Task Engineer.
- Weather monitoring - closer monitoring of prevailing weather conditions.
- Readiness review meeting - involving all involved personnel to ensure all risks are considered and adequate controls implemented.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

MIDDLE EAST ONSHORE

DATE: 05 Dec 2023
COUNTRY: Kurdistan Region Of Iraq
FUNCTION: Unspecified
CAUSE: Unspecified - Other
ACTIVITY: Transport - Land
RULE: Driving

NARRATIVE:

A 3rd party driver, while driving on a straight section of the road, lost control of the vehicle due to using his cell phone to access social media. He braked hard, causing the vehicle to skid and roll over, narrowly avoiding a tree. This resulted in slight damage to the vehicle, a small diesel spill, and minor injuries to the driver. The scene was made safe traffic controlled; the driver was checked over by the Medic and evacuated to hospital. The truck was righted by two front loaders and removed from the scene. Clean-up was conducted by the spill response team. The driver was not wearing a seatbelt so the injuries could have been much worse. The contractor brought new trucks and drivers at the end of the project to speed up the finish without onboarding the new hires.

WHAT WENT WRONG:

Third-party driver lost control of the vehicle while using a cell phone to access social media.

Driver's distraction led to hard braking, causing the vehicle to skid and roll over.

Lack of adherence to safety protocols, including failure to wear a seatbelt, increased the severity of injuries.

Contractor's decision to bring in new trucks and drivers without proper onboarding potentially contributed to the incident by introducing inexperienced personnel.

Failure to equip all vehicles with In-Vehicle Monitoring Systems (IVMS) on company projects.

Inadequate enforcement of a 'no mobile phone use policy' while driving.

Insufficient periodic assessments of third-party drivers to ensure compliance with safety standards and necessary qualifications.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Most accidents tend to occur towards the end of the day, shift, rotation, or project, as individuals may become fatigued and let down their guard. Vigilance should be maintained throughout.
- Small changes in behaviour, such as using a cell phone while driving, can have significant impacts on safety if not evaluated and managed effectively.
- Safety rules must be straightforward, clearly communicated, consistently enforced, and accompanied by consequences for infractions to be effective.
- Assumptions regarding deviations from safety protocols should not be made unless formal sign-off has been obtained, highlighting the importance of clear communication and documentation.
- Land transport remains a significant risk in the oil and gas industry, emphasizing the critical need for continuous improvement in safety measures and practices to prevent accidents with the potential for severe consequences.

MIDDLE EAST ONSHORE

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Work or motion at improper speed

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

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DATE: 20 Jul 2023
COUNTRY: Oman
FUNCTION: Exploration
CAUSE: Struck by (not dropped object)
ACTIVITY: Transport - Land
RULE: Other issue – no applicable rule

NARRATIVE:

A contractor GPS operator hit his head against an Antenna metal bar which is fixed on top of the vehicle while getting out of his stuck vehicle. IP was mobilized to base camp clinic for further diagnosis and examination. He was examined by crew doctor, three stiches were applied on the affected area.

WHAT WENT WRONG:

- Improper decision making or lack of judgement - driver did not assess the sand/terrain condition to avoid the vehicle becoming stuck in soft sand.
- Distracted by other concerns - driver was distracted due to vehicle getting stuck.
- Inattention to footing and surroundings – driver didn't notice the change of the antenna height after the vehicle became stuck and tire deflation.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Awareness about Personnel attention/behaviour during work.
- R10 Antenna holder to be secured with foam protective rolls.
- Antenna Holder to be more visible by reflective tape.
- Personnel awareness (don't panic or get frustrated) when stuck or broken down.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

MIDDLE EAST ONSHORE

DATE: 24 Nov 2023
COUNTRY: Oman
FUNCTION: Drilling
CAUSE: Unspecified - Other
ACTIVITY: Drilling, workover, well operations
RULE: Other issue – no applicable rule

NARRATIVE:

At one of the rigs sites, it was found that mud pump isolated without tag and only one padlock.

WHAT WENT WRONG:

- Supervisors became complacent to the task.
- Daily Reviews of the PTW Process to ensure compliance were not implemented.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Daily review with all parties involved with the process is required – do not become complacent.
- A yearly review of the process is required to ensure best practices are kept up.
- Only Issuing Authority and CEs allowed to isolate in SCR.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 07 Aug 2023
COUNTRY: Oman
FUNCTION: Drilling
CAUSE: Unspecified - Other
ACTIVITY: Drilling, workover, well operations
RULE: Other issue – no applicable rule

NARRATIVE:

During P/U and RIH 8.5" directional BHA at depth 500m, the upper pipe arm did not fully retract after putting the pipe in the mouse hole, the upper arm moved up by DP then got released. The upper arm sling got cut and the upper arm fell on the lower pipe arm and damaged it.

WHAT WENT WRONG:

- Upper arm did not retract.
- Drill Pipe Handler Monitor get alarm (Warning alarm).
- The pipe handler operator didn't pay attention while operating the pipe handler control panel.
- Poor communication and coordination.
- Lake of supervision.

MIDDLE EAST ONSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Ensure to use SWA if you see unsafe conditions or unsafe acts. Pay attention while working on the pipe handler control panel.
- Ensure the antifelting device is activated.
- Ensure the upper pipe arm function properly and is inspected before starting the job.
- Ensure proper communication and coordination while working.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

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DATE: 27 Jun 2023
COUNTRY: Oman
FUNCTION: Unspecified
CAUSE: Slips and trips (at same height)
ACTIVITY: Transport - Land
RULE: Other issue – no applicable rule

NARRATIVE:

While offloading sweet water at one of the camps, the tanker driver moved backward, where his left foot slipped on a stone behind which caused a twist to the foot, resulting in swelling of his foot.

WHAT WENT WRONG:

- Driver did not notice that a stone was behind him.
- Driver moved backward before insuring safe surrounding.
- Lack of housekeeping at the water filling area.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Always Check Surroundings: It is essential to be vigilant and check the surroundings before any movement.
- Ensure Clear Working Area: Before commencing any task, ensure that the ground in the working area is clear and free of any objects that might pose a risk.
- Remove Hard Objects: Before moving backwards, remove or mark any hard objects or obstacles to prevent accidents and injuries.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Organizational: Inadequate communication

MIDDLE EAST ONSHORE

DATE: 20 Mar 2023
COUNTRY: Qatar
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Maintenance, inspection, testing
RULE: Work authorization

NARRATIVE:

On 20 March 2023, a leak was identified coming from a connection hub on the platform. The leak was monitored, and an Initial Incident Notification (IIN) was raised seven days later, on 27 March. Attempts were made to stop the leak on 10 April by torqueing the stud bolts. After torqueing the stud bolts, the leak continued to persist. Another attempt to arrest the leak was made on 22 April, using a new seal ring, stud bolts, and nuts. A decision was made not to use a new blind which was available on site. Even after the second attempt to address the leak, the leak continued. On 27 April, a third intervention was made to stop the leak using the new seal ring that was used on the 22 April attempt, new OEM studs and bolts, and a new clamp. After the third attempt, the leak was stopped.

WHAT WENT WRONG:

Incident resulted from maintenance activities not having the original spare parts available during replacement.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Execution team should have engaged OEM during the planning stage and proactively ordered the complete set of spares, knowing that this was a critical path activity.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

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DATE: 03 Apr 2023
COUNTRY: Qatar
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Office, warehouse, accommodation, catering
RULE: Unspecified

NARRATIVE:

A Contractor was performing his routine activities with his personal mobile phone, in his front pocket. The Contractor noticed the mobile phone was increasing in temperature, becoming uncomfortably hot. The Contractor removed the phone from his pocket and, due to the heat, threw the phone away. The phone landed on the floor and exploded, burning the carpet, and generating smoke, which triggered the smoke alarm in the Fire Alarm Control Panel (FACP).

After receiving the alarm, Company security investigated the location of the alarm and identified the source of smoke from the damaged phone.

WHAT WENT WRONG:

Potential manufacturing defect of electronic devices.

MIDDLE EAST ONSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Review the brand before buying any phone.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 13 Mar 2023
COUNTRY: Qatar
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Office, warehouse, accommodation, catering
RULE: Unspecified

NARRATIVE:

Exhaust fan cover fell from North Warehouse Rooftop.

WHAT WENT WRONG:

- Wear and tear: Failure was due to vibration, material corrosion, and aging.
- Service life: Fan covers were not replaced after the expected service life (15 years) of roof-mounted exhaust fans.
- Inadequate access: There is no access to the roof fans for inspection or maintenance.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Replacement of in-service roof exhaust fans beyond its designed service life.
- Develop a planned inspection and replacement strategy for equipment/fixtures mounted on the roof.
- Report any potential falling objects immediately.

DATE: 02 Feb 2023
COUNTRY: Qatar
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Production operations
RULE: Other issue – no applicable rule

NARRATIVE:

At 10:27 am, an LOPC of toxic gases took place in Fire Zone activating PAGA and a full plant mustering. Approximately 1602 personnel evacuated and mustered with no injuries or exposure of any personnel to the released gases.

The incident was classified as high potential due to H2S exceeding the IDLH limit of 100 ppmv. It is worth noting that the incident manifested when stream 1 flaring was ongoing during which the drum pressure increased to 0.39 barg from its normal operating pressure of 0.02 barg since it floats with the flare system. The toxic gas detector activated was 1300GDT24D118 recording a value of 100 ppmv. A slight reaction of 1300GDT24D104 was also detected reading a value of 2 ppmv. Wind direction plays a vital role in which detector activates and what levels of H2S they will read.

MIDDLE EAST ONSHORE

WHAT WENT WRONG:

- Facility Management of Change.
- TGTU bypass line not in dead-leg programme.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

All failed pipe spools to be sent for 3rd party metallurgical investigation.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 28 Jul 2023
COUNTRY: Qatar
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Production operations
RULE: Unspecified

NARRATIVE:

Elevator door, fell to the ground from 26 metres high.

WHAT WENT WRONG:

Design of the elevator door closing mechanism is not user friendly. Such design makes it difficult for non-familiar users to operate the door and result in application of excessive force that subsequently damages the door operation mechanism.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Elevator door mechanism design requires upgrade to maintain long term integrity.

Similar elevator door failures have happened in the past in other assets, however, lessons were not applied across the site.

Do not use elevators unless you are trained to use them and know what to do in case of an emergency.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

MIDDLE EAST ONSHORE

DATE: 02 Apr 2023
COUNTRY: Qatar
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Production operations
RULE: Unspecified

NARRATIVE:

A loading hook fell from the davit arm of a Flue Gas incinerator.

WHAT WENT WRONG:

- Sustain Barriers: The Davit Arm was not used after the handover from commissioning to Operations, therefore it was not included in the equipment maintenance strategy.
- Apply Procedure: The handover process was not followed correctly. The Davit Arm's functional location was missed therefore the equipment did not undergo any inspections.
- Watch for Weak Signals: The loading hook accessory attached to the Davit Arm was in a corroded state.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Ensure lifting accessories which are not undergoing bi-annual and/or annual inspection are removed from the lifting device to eliminate the potential falling object hazard and stored properly by Rigging team to prevent them from getting damaged or worn out. Ensure prior to handover process as per procedure that Functional Locations for Lifting Equipment are created to develop Equipment Strategy.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: 27 Sep 2023
COUNTRY: Qatar
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Production operations
RULE: Unspecified

NARRATIVE:

A hoist hook fell from RGC-2 bypass stack David arm.

WHAT WENT WRONG:

Sustain Barriers: The Davit Arm was not used after the handover from commissioning to Operations, therefore it was not included in the equipment maintenance strategy. A very similar incident happened previously, however Corrective Actions Recommendations (CARs) were not completed and the incident repeated.

- Apply Procedure: The handover process was not followed correctly. The Davit Arm's functional location was missed therefore the equipment did not undergo any inspections.
- Watch for Weak Signal: The loading hook accessory attached to the davit arm was corroded.

MIDDLE EAST ONSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Follow up on the timely implementation of corrective actions for this incident and previous similar incidents to ensure they have been implemented correctly.

Davit Arm accessories e.g., hook blocks/chain blocks etc. are to be removed and kept with Support Services (Rigging Team).

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: 10 Oct 2023
COUNTRY: Qatar
FUNCTION: Production
CAUSE: Water related, drowning
ACTIVITY: Diving (incl. decompression), subsea, ROV
RULE: Other issue – no applicable rule

NARRATIVE:

Men overboard incident: While preparing for a dive on a Diving Support Vessel's workboat for an intake cleaning operation, the weather changed and sea conditions deteriorated rapidly. The dive was suspended, but the vessel took on water and turned temporarily on its side, throwing the crew of 6 persons into the water. Five crew members swam to the boat landing, and one crew member swam to a life ring that was thrown to him from the platform. The incident resulted in one first aid case.

WHAT WENT WRONG:

Water ingressed into the vessel. The Diver was recovered but in the process, the vessel took on more water. The single auto-start bilge pump failed to function, causing the aft part of the vessel to become partially submerged. The free door at the diver door is too low, allowing additional water to enter the vessel in marginal weather conditions. Flooding of the vessel's engines rendered them inoperable. Weather conditions at the start were acceptable, but rapidly deteriorated.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Modifications to the diver door freeboard to increase height.
- New diver door to be fabricated and installed.
- Replace bilge switch/re-install bilge pump/install back-up bilge pump.
- Full inspection of all mechanical and electrical systems and testing.
- Pay closer attention to weather forecasts before embarking on tasks.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

MIDDLE EAST ONSHORE

DATE: **15 Mar 2023**
COUNTRY: **Qatar**
FUNCTION: Construction
CAUSE: Dropped objects
ACTIVITY: Construction, commissioning, decommissioning
RULE: Unspecified

NARRATIVE:

A scaffold coupler clamp dropped from +1 platform and fell 20m, landing on the annular area.

WHAT WENT WRONG:

The double clamp was installed incorrectly and the scaffolder was unaware it was improperly fastened.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Enhance current 'Verification of Competency' performed with Craft Training for all scaffold personnel.
- Communicate the SIMOPS information such as daily SIMOPS coordination meeting outcome, exclusion zones, etc. to workers.
- An exclusion zone/safe work platform diagram shall be posted at all access staircases.
- Work at elevated areas shall not be commenced if the dropped object prevention requirements are not complied with.
- Assign a watchman to enforce the established exclusion/drop zone at the roof level to ensure no personnel enter the exclusion/drop zone.
- A Permit to Work (PTW) shall not be issued/revalidated if the controls required for the work are not implemented and in place.
- Disciplinary action shall be taken on PTW issuer and receiver for not implementing the PTW requirements.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: **23 Mar 2023**
COUNTRY: **Qatar**
FUNCTION: Construction
CAUSE: Dropped objects
ACTIVITY: Construction, commissioning, decommissioning
RULE: Unspecified

NARRATIVE:

TMS Platform Climbing Cone weighing 2.3kg dropped and fell 15 metres.

WHAT WENT WRONG:

Housekeeping activities were not supervised and controlled correctly, material storage box was full of removed climbing cones, and the green mesh to prevent potential dropped objects was not installed correctly.

MIDDLE EAST ONSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Establish exclusion zones and barricades before commencing work at elevated areas with the potential for dropped objects.
- Housekeeping is part of work activities and should be performed regularly to prevent potential dropped/falling objects.
- Work on elevated work platforms shall not commence if dropped object prevention requirements are not fully complied with.
- Entry into an exclusion/drop zone is to be enforced and controlled so that there is no unauthorized entry.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 29 Jul 2023
COUNTRY: Qatar
FUNCTION: Construction
CAUSE: Dropped objects
ACTIVITY: Construction, commissioning, decommissioning
RULE: Unspecified

NARRATIVE:

Timber with tie-rod fell 6 metres.

WHAT WENT WRONG:

The process of installing and removal of tie-rod was not captured in the MS and JSA for the BOP.

The Supervisor and chargehand were not aware of the safe process of removal of tie-rod from the slab.

The MS and JSA for the Concrete Works for BOP were not updated since July 2022.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Correct sequence of installation and dismantling of tie-rods assembly should be ensured to avoid dropped event.
- Establish and enforce an exclusion/drop zone with adequate barricade when drop potential activities are undertaken.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

MIDDLE EAST ONSHORE

DATE: 28 Aug 2023
COUNTRY: Qatar
FUNCTION: Construction
CAUSE: Exposure electrical
ACTIVITY: Excavation, trenching, ground disturbance
RULE: Energy isolation

NARRATIVE:

During mechanical excavation to create an entry pit hole for horizontal directional drilling aimed at installing a new 6" flowline across a road, the excavator damaged an unidentified live 11kV underground power cable supplying electricity to a road lighting substation at a main pump station. Although there were no injuries, the live cable damage caused power supply interruption, voltage dip, and trip of several equipment, and power outage across many areas including a residential area. Furthermore, people assigned to investigate the damage caused by the excavator heard arc blast from inside of the trench. This created further unsafe conditions which could potentially have caused electrocution and fatalities.

WHAT WENT WRONG:

- Ineffective Permit to work: Mechanical excavation was performed outside the authorized and scanned area.
- GIS Mmp has not been updated with all underground utilities.
- Lack of underground and above-ground utilities warning signs to identify the 11 KV cable which was laid long ago.
- Protection tiles and sand bedding overlooked during the mechanical excavation.
- Ineffective communication between the Contractor and the PTW authorities regarding accurate area to be excavated.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Need to enhance implementation of the existing PTW system.
- Construction team to review the site survey and construction area to ensure proper safety controls are in place for safe execution of HDD work activities.
- Install identification markers for all underground utilities.
- Relevant Operations Engineering Drawing Office representative to visit worksites and visually verify existing conditions prior to endorsement of the Excavation Certificates.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Organizational: Inadequate communication

MIDDLE EAST ONSHORE

DATE: 31 Dec 2023
COUNTRY: Qatar
FUNCTION: Construction
CAUSE: Falls from height
ACTIVITY: Construction, commissioning, decommissioning
RULE: Unspecified

NARRATIVE:

Structural steel beam fell from a height of 2m.

WHAT WENT WRONG:

The rigging team rigged and lifted four (4) pieces of structural template beams (approx. 3m, 40kg each) together. When the load was suspended approximately two metres into the air, one of the structural steel frames slipped from the rigging and fell to the ground landing inside the lifting barricaded area.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Make sure that all loads are adequately secured before lifting by crane.
- Inspect all lifting tools and equipment before use.
- Conduct an LMRA and apply adequate controls before every lift.
- Properly centre all loads and place slings so that the load is secured.
- Never walk or work under a suspended load.
- LSR – Walking under Suspended load shall be complied with at all times.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: 08 Jan 2023
COUNTRY: Qatar
FUNCTION: Construction
CAUSE: Unspecified - Other
ACTIVITY: Construction, commissioning, decommissioning
RULE: Unspecified

NARRATIVE:

Falling object, a bolt head fell from 4th level of tower crane mast.

WHAT WENT WRONG:

Crane was left unattended during night shift and it experienced multiple emergency stops.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Regular checklists verification and inspection of crane.

MIDDLE EAST ONSHORE

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)

DATE: 16 Oct 2023
COUNTRY: Qatar
FUNCTION: Construction
CAUSE: Unspecified - Other
ACTIVITY: Construction, commissioning, decommissioning
RULE: Unspecified

NARRATIVE:

Chain block hook caught on the bottom of the structural beam and released suddenly.

WHAT WENT WRONG:

While hoisting the pipe spool, the chain block hook caught at the bottom of the structural beam, and as the lift progressed, tension built up in the sling and chain block, and eventually the caught-up chain block hook opened and got released from the stuck position. When released, the chain block sprung up and hit the upper section of the beam. The chain block hook was damaged due to impact and tension.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Ensure all unnecessary tools and equipment that are not part of the rigging process must be removed/secured prior to the lifting.
- Always follow the lifting plan and supervisor shall ensure the rigging arrangements are in compliance with plan.
- Lifting is one of the critical activities, therefore ensure safe system/sequence of work is followed to avoid any series event.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)

DATE: 14 Nov 2023
COUNTRY: Qatar
FUNCTION: Construction
CAUSE: Unspecified - Other
ACTIVITY: Construction, commissioning, decommissioning
RULE: Unspecified

NARRATIVE:

Forklift moved backwards on a trailer bed and damaged the trailer hydraulic ramp.

WHAT WENT WRONG:

It appears that the forklift was not properly secured with a ratchet and lever chain arrangement, and no chock blocks were used to immobilize the forklift.

MIDDLE EAST ONSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- After loading heavy equipment on a lowbed trailer, chock the wheels and then secure the load/equipment with chain load binders/lashing arrangement.
- Do not commence transportation without an escort when it is required.
- Do not commence transportation without securing the load.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

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DATE: **28 Feb 2023**
COUNTRY: **Qatar**
FUNCTION: Unspecified
CAUSE: Exposure electrical
ACTIVITY: Office, warehouse, accommodation, catering
RULE: Energy isolation

NARRATIVE:

As a part of the office revamping work, a technician was discharging water from the sprinkler line. Surprisingly, an exposed electrical cable (tagged as NO POWER) contacted the ceiling metal frame and generated sparks. The scene was in proximity of the other technician who was standing on the podium ladder.

WHAT WENT WRONG:

- Lockout and Tagout (LOTO) system was not followed.
- Temporary connection was looped with existing live connection.
- Inadequate supervision and no formal handover process between contracting teams.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- LOTO supervision was reinforced.
- Implemented rule for the Permit to Work to be counter signed by Company Safety Officers after verifying all the control measures are in place, including electrical isolation.
- Introduced formal shift handover sheet that will be used between shifts.
- Installed Temporary Distribution Board (DB) by contractor. The DB are inspected and tagged by Electrical Engineer.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

MIDDLE EAST ONSHORE

DATE: 05 Jun 2023
COUNTRY: UAE
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Maintenance, inspection, testing
RULE: Line of fire

NARRATIVE:

At 11:34 hrs, a field operator reported that, after replacing of TRF CRS Filter elements, and while trying to close the Filter door, it fell on the ground.

WHAT WENT WRONG:

Crevice corrosion and scaling due to the ingress of moisture.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Prioritization of SAP to be reviewed.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

MIDDLE EAST OFFSHORE

DATE: 23 Feb 2023
COUNTRY: Qatar
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well operations
RULE: Safe mechanical lifting

NARRATIVE:

A crew member and 3rd party personnel were trying to lower a 3" Wilden Pump (weight: 75kg) using the handrail as an anchor point and 0.5" manila rope as a lifting aid. In the process of lowering down the Wilden Pump, the manila rope parted at approximate 1.2m from the deck causing the Wilden Pump to drop to the deck. The Wilden Pump was lowered to the deck below in order to overcome loss of suction issue when pumping MUD from Intermediate Bulk Container (IBC) tank at main deck level to the shaker flow line (one deck level above IBC tank).

WHAT WENT WRONG:

- Use of non-certified rope for lifting 75kg equipment.
- Poor risk assessment: Failure to intervene/Stop work authority.
- No permit to work used (Work at height).
- No lesson learned from previous incidents.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- All drilling contractors and service companies shall complete the Risk assessment and Permit to Work (PTW) training.
- Ban the use of non-certified ropes for lifting on all drilling rigs. PTW to ensure the safe use of certified ropes.
- Review the design of the Mud transfer to avoid moving Wilden pump around.
- Management of Change awareness training and Manual handling training performed for all parties involved in the incident.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

MIDDLE EAST OFFSHORE

DATE: **22 Mar 2023**
COUNTRY: **Qatar**
FUNCTION: Drilling
CAUSE: Pressure release
ACTIVITY: Drilling, workover, well operations
RULE: Energy isolation

NARRATIVE:

Drill crew was breaking out a stand of 5 inch Drill-Pipe. Having set the slips and installed the Mud Bucket around the connection, the Floormen moved out of the red zone. The Driller continued breaking out the connection with Iron Roughneck and Top Drive System. Having broken out the connection there was a sudden rush of drilling mud at the tool joint and pushed the mud bucket upwards whilst under mud pump pressure (1800 psi).

WHAT WENT WRONG:

Inadequate Job safety analysis/specific instruction to ensure that appropriate steps are taken to confirm Mud Pumps are switched off before any connections or disconnections made.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Review the Job safety analysis and the Standard operating practice for backreaming operation.
- Consider the use of secondary pressure gauge installed on drill floor where rig crew can observe the pressure reading and confirm zero pressure to the driller.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: **27 May 2023**
COUNTRY: **Qatar**
FUNCTION: Drilling
CAUSE: Unspecified - Other
ACTIVITY: Drilling, workover, well operations
RULE: Unspecified

NARRATIVE:

Uncontrolled release of torque resulted in cement head destabilization.

WHAT WENT WRONG:

The TDS stalled because command communication (an integrated drilling control system) was lost, and there was no scheduled Preventive Maintenance (PM) to back up or dump data on the Compact Flash (CF) card to prevent communication loss when the card becomes full.

The Driller was unaware that RPM and torque values reset to zero automatically when TDS command communication is lost.

MIDDLE EAST OFFSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Implement a PM programme to clear excess data from the CF card effectively.
- Ensure the Driller knows how to restart TDS if it trips with torque in the string.
- Properly assess third party or service provider equipment in the drill string to withstand uncontrolled string rotation driven by TDS. Implement Red Zone management based on dropped object risk assessments.

DATE: 02 Apr 2023
COUNTRY: Qatar
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Production operations
RULE: Unspecified

NARRATIVE:

A section of HVAC Duct (15kg) fell from a height of 5m to level 4 deck.

WHAT WENT WRONG:

The undersized rivets used for duct cladding did not meet the specifications in Thermal Insulation Standard RG S 00 1390 001, which made them a weak point and reduced their durability when exposed to oxidation.

The sealant and rivet size used in the location were not suitable for the conditions.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Outdoor HVAC systems are exposed to high winds and corrosion. Ensure securement of cladding meets the required specifications.
- A full survey of HVAC cladding and related structures was conducted based on Thermal Insulation Standard RG S 00 1390 001.
- Standard sealant is to be re-applied, and the rivets are to be replaced, as per specification in Thermal Insulation Standard RG S 00 1390 001.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

MIDDLE EAST OFFSHORE

DATE: 07 Aug 2023
COUNTRY: Qatar
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Production operations
RULE: Unspecified

NARRATIVE:

Monkey ladder top support weighing 10kg fell 7 metres.

WHAT WENT WRONG:

FOP Apply Procedures: The ladder had been exposed to a marine environment for years without inspection or maintenance.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- During routine rounds, observe the condition of ladders and report any defects for maintenance.
- Before using any ladder, visually check for integrity and avoid using if any defects are found.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 28 Aug 2023
COUNTRY: Qatar
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Unspecified - other
RULE: Bypassing safety controls

NARRATIVE:

During bunkering from the supply vessel into the Platform Diesel Storage Tank, a large volume of diesel leaking from the Weather deck onto the main deck was observed and reported. The source of the leak was coming from the drains on the cellar deck.

WHAT WENT WRONG:

- Central control room operator changed the Level Alarm setpoint.
- Level transmitter was not calibrated.
- Klaxon alarm at boat landing was not working.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Site standing instruction to be issued and signed prior to commencing the bunkering procedure.
- Recalibrate the Level Transmitter.
- Replace Klaxon alarm device at boat landing as per P&ID.
- Create and update diesel bunkering procedure to ensure functionality of trip settings and alarms prior to diesel bunkering.

MIDDLE EAST OFFSHORE

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: 25 Oct 2023
COUNTRY: Qatar
FUNCTION: Construction
CAUSE: Exposure noise, chemical, biological, vibration, extreme temperature
ACTIVITY: Maintenance, inspection, testing
RULE: Work authorization

NARRATIVE:

For J-Tube inspection using manual ultrasound test, 8 divers from the diving vessel approached the platform in a small dive boat. The work scope was located at the splash zone of the boat landing of the platform. On arrival at the boat landing, all divers' personal toxic clips started flashing/alarming. Dive team immediately returned to diving vessel and re-boarded vessel with divers experiencing symptoms.

WHAT WENT WRONG:

- Improper job preparation and Permit to Work.
- Improper Risk assessment.
- Inappropriate mitigation measures for the downgraded situation.
- Back pressure due to blocked flame arrestor.
- Temporary 2" vent line.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Ensure all supervisors are trained in Permit to Work and Management of downgraded Situation.
- Ensure the involvement of all parties at all stages of Risk Assessment and live review and consider all the live downgraded situations.
- Look for the next available shutdown opportunity to repair and replace the defective equipment.
- Rectify the design deficiency of flame arrestor.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

MIDDLE EAST OFFSHORE

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing
PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)
PROCESS (CONDITIONS): Organizational: Inadequate training/competence
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment
PROCESS (CONDITIONS): Organizational: Inadequate communication
PROCESS (CONDITIONS): Organizational: Inadequate supervision
PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

NORTH AMERICA ONSHORE

DATE: **10 Jan 2023**
COUNTRY: **Canada**
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well operations
RULE: Line of fire

NARRATIVE:

A wireline crew was on location to complete a rigless downhole abandonment. The wire line crew asked a combo-vac operator to operate the winch controls that would lift the next section of the bailer out of the hole. When the combo-vac operator pulled the lever they were shown, the mast extended instead of retracting the winch, causing the wing chile to crown-out. The winch hook, headache ball, pack-off head, and Bowen union crossover assembly fell from the top of the mast, striking a wireline crew member standing at the wellhead, breaking their forearm.

WHAT WENT WRONG:

The wireline crew asked another individual to operate the controls of the wireline equipment. The individual did not have any experience operating that type of equipment.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Do not operate equipment you are not qualified to operate.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

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DATE: **20 Aug 2023**
COUNTRY: **Canada**
FUNCTION: Drilling
CAUSE: Falls from height
ACTIVITY: Drilling, workover, well operations
RULE: Working at height

NARRATIVE:

At 7:50am a member of the rig crew fell from the BOP walk around in the sub base of the rig. While working on the BOP platform, installing a new "Katch Kan" tarp, a worker contacted a handrail. The worker was attempting to move a magnet to tighten the tarp. A failure of the handrail occurred and the worker fell 4.85m to the floor below. During the fall, the worker contacted the main I-beam and landed supine on the floor next to the cellar. All operations have been suspended since the event. Company HESR contacted and reported event to OH&S. OH&S team arrived on location at 6:00 pm and conducted investigations until 10:30 pm. A stop work order was issued specifically for work on the BOP platform and does not prevent moving forward with casing running operations. Safety stand down held from 10:30 - 11:30 pm. Operations resumed at 11:30 pm.

WHAT WENT WRONG:

Design standards not utilized. Inadequate handrails.

NORTH AMERICA ONSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

The key lessons learned from this incident are:

- Removable handrails should have pins in all sockets, not just one. They should also be designed with mechanical attachments to adjacent handrails to provide for redundant support.
- Removable aluminium handrails and decks cannot be treated like carbon steel for weld repairs. This knowledge is likely difficult to retain as incident memory fades and new workers are brought in. Safeguards (in this case stickers and stencilling with recurring inspection to ensure they stay in place) need to be in place to ensure this is not forgotten.
- Handrail and deck inspections must be specific enough to ensure highest failure risk areas and components are inspected on a recurring basis. Design and repair mistakes and equipment degradation can happen. Recurring, specific inspections must be in place as a final safeguard to ensure, when it happens, these are identified.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

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DATE: 19 Sep 2023
COUNTRY: Canada
FUNCTION: Drilling
CAUSE: Struck by (not dropped object)
ACTIVITY: Lifting, crane, rigging, deck operations
RULE: Line of fire

NARRATIVE:

Snubbing crew members were in the process of preparing to raise and stow an Extension Leg. The Operator asked the IP to shake the Extension Leg Support Cable to make it easier to hammer out the Extension Leg Primary Pin. The IP grabbed the cable with his Right Hand and braced his Left Hand on the edge of the Extension Leg for balance. While shaking the Extension Leg Cable, the Primary Pin was hammered out. This caused the Extension Leg Sleeve to unexpectantly slide down the Extension Leg due to a missing Secondary Pin, resulting in partial amputation of left thumb.

WHAT WENT WRONG:

Missing secondary pin caused the extension leg sleeve to unexpectantly slide down the extension leg.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Not available.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

NORTH AMERICA ONSHORE

DATE: 06 Sep 2023
COUNTRY: Canada
FUNCTION: Production
CAUSE: Exposure noise, chemical, biological, vibration, extreme temperature
ACTIVITY: Production operations
RULE: Unspecified

NARRATIVE:

Tubing drain line split open, releasing H2S gas into the high-pressure inlet building. Operators isolated and locked out the site, drained all inlets and scrubbers of fluid. Operations left the tubing drain line off the belly drain open. The site was being shut in with perimeter blinds and inlet V110 belly drain was open to drain remaining fluid from the vessel. Operators walked over to the sales line to open the flare to bleed off the pressure in the facility. While walking back over to the inlet to open the flare line, operator noticed gas coming from the building.

WHAT WENT WRONG:

Tubing drain line split open, releasing H2S gas into the high-pressure inlet building.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Move H2S visual display from inside to outside of all buildings.
- Work with engineering to minimize or eliminate the amount of instrumentation tubing used on the bridle and sight glass drain systems.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 04 Aug 2023
COUNTRY: Canada
FUNCTION: Production
CAUSE: Exposure noise, chemical, biological, vibration, extreme temperature
ACTIVITY: Production operations
RULE: Unspecified

NARRATIVE:

Security guard noticed that one of the wells was visually leaking. Security notified Operations and the pad Emergency Shut Down (ESD) was triggered when they arrived on scene. After the pad was ESD, Operations donned SCBA and attempted to shut in the well.

WHAT WENT WRONG:

Well Leaking.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Create PM programme for maintenance of surface ESDs.
- Implement the Well Integrity programme and Inspection, Testing and Maintenance process for area.

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CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: **14 Dec 2023**
COUNTRY: **Canada**
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Production operations
RULE: Working at height

NARRATIVE:

Contractor's worker was using a personnel basket to connect slings to lifting lugs while setting a 400 bbl tank. The worker was observed on top of the tank, outside the personnel basket without being tied off. The worker was approximately 30 feet from ground level. No injury was sustained.

WHAT WENT WRONG:

The contractor's worker was unaware of company policy which prohibits exiting man baskets onto the top of tanks without tying off, which they understood to be a commonly accepted practice at other companies' facilities.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Ensure that orientation of contractors' workers clearly communicates company rules, and those rules are understood and followed by the contractors' workers.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: **23 Nov 2023**
COUNTRY: **Canada**
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Production operations
RULE: Unspecified

NARRATIVE:

A well fire was reported by a member of the public. Upon responding, operations confirmed that a NGL pipeline had failed and was on fire. The line was isolated and the fire self-extinguished when the line de-pressured. Prior to the fire, operations had already been en-route to the site to perform work at the location of the failure, but had been delayed by a phone call. No injuries were sustained.

WHAT WENT WRONG:

Pipeline integrity was compromised by a tether during the inline inspection. Debris observed during the inspection was not initially recognized as a serious threat to pipeline integrity and information about the debris was not

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communicated to the Operations group prior to start-up. The multiple bends on the pipeline were not accounted for during the work planning and hazard assessment phase of the inline inspection.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Drawings/prints should be up to date for work planning processes. Turnover processes should be formalized to ensure all necessary information is communicated.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: 25 Nov 2023
COUNTRY: Canada
FUNCTION: Unspecified
CAUSE: Struck by (not dropped object)
ACTIVITY: Diving (incl. decompression), subsea, ROV
RULE: Driving

NARRATIVE:

Vehicle struck walking worker in the lower back with bumper.

WHAT WENT WRONG:

Workers failed to follow company safe operating procedures. Worker struck did not wear High Visibility Clothing.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Operation has relocated garbage bin to an area within the Laydown Yard that alleviates congestion and allows workers to transfer garbage from their vehicles into the bin without having to back into a congested area.
- Backup alarms are to be installed on all vehicles.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Personal Protective Equipment not used or used improperly

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 08 Aug 2023
COUNTRY: USA
FUNCTION: Drilling
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Drilling, workover, well operations
RULE: Line of fire

NARRATIVE:

Supervisor (IP) and team were attempting to secure a 2162lbs block with two U-bolts to a support stand. The clearance between the stand and the ground was too tight to make-up the nuts on the U-Bolts. A forklift was used to pick up the stand with block using a single sling cradled through two hooks on the bottom of one of the forks and

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wrapped under the stand height adjust cranking points on the stand. When the stand was 2 feet from the ground, the forklift held the load while IP and 2 teammates commenced making up the nuts on the U-bolts. The IP applied a wrench to work a nut on one corner of the load. This action seems to have caused the slings to slide through the hooks towards him. The centre of gravity was off sufficiently that the block tumbled from the stand.

WHAT WENT WRONG:

Potential consequences not understood.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Team to develop a documented plan for rig/pad moves for all Wells operations.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

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DATE: 08 Mar 2023
COUNTRY: USA
FUNCTION: Drilling
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Drilling, workover, well operations
RULE: Safe mechanical lifting

NARRATIVE:

Rig crew began rigging down the wellhead flange. The rig team member placed his left hand under the suspended flange attempting to put a cable on the ring gasket to hold it up to install the slips. The tubing collar pulled off the B-1 flange, trapping the worker's hand between the flange and the wellhead.

WHAT WENT WRONG:

Adapter flange and pup joint thread failure.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Implement a hands-free device for accessing ring gasket.
- Plate has been constructed to use for setting of slips without having to move seal ring to place slips.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: **28 Jul 2023**
COUNTRY: **USA**
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well operations
RULE: Safe mechanical lifting

NARRATIVE:

Rig move Truck Pusher was spotting for a haul truck when a 47lbs pit sensor hole cover rolled off the top of the mud pit skid falling 7.8ft, hitting him in the left shoulder.

WHAT WENT WRONG:

- Suction pit was moved from its installed position prior to being fully secured and without clear authorization from the Rig Contractor that it was ready to move.
- Red Zone criteria for Trucking personnel (Truck Pushers/Swampers) when spotting equipment not clearly defined.
- Rig Contractor personnel did not verify that the pit sensor hole cover secondary retention cable was installed pre-rig move and no periodic inspection exists.
- Pit sensor hole cover is a legacy design that is oversized compared to the new sensors it covers and heavier than it needs to be.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Rig Contractors to define clear custody transfer practice for transit, providing clear authorization that a load is secured and ready. Rig Movers to require a confirmation/verification of ready to move before picking up any load.
- Trucking Company to develop a clear Red Zone policy and train employees on recognizing line of fire hazards during the move. Policy to include periodic HSE field verifications of safeguard understanding and integrity.
- Trucking Company to develop Rig Move pre-job briefing documentation for Red Zone practices considering applicable Life-Saving Rules.
- Rig Contractor to inspect all pit sensor hole cover secondary retention cables and replace any missing or damaged components.
- Rig Contractor to complete engineering design review of pit sensor hole cover for contracted rig types.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

NORTH AMERICA ONSHORE

DATE: **22 Jun 2023**
COUNTRY: **USA**
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
RULE: Safe mechanical lifting

NARRATIVE:

While offloading a riser from a supply vessel to the rig, the riser began to rotate unexpectedly, and the crane operator needed to stabilize the load. The crane operator set the riser down in the riser bay against the stanchions to stabilize the load. When the crane operator picked the riser up, the back end of the riser became caught on the stanchion at the flange. The crane operator began to manoeuvre the load inboard when the rig took a roll and the riser freed from the stanchion and contacted a 6 5/8" BOP Test Joint that was secured vertically against the Riser Skate Catwalk. This contact allowed the BOP Test Joint to come free from the securement resulting in the top leaning outboard and the bottom to slide inboard. The Test Joint passed through the rigging loft under the riser skate and came to rest on the deck that was identified as the Safe Step Back Area per the Job By Design. The Test Joint fell approximately 14.5 feet and weighed 4,546.2 pounds.

WHAT WENT WRONG:

Investigation identified that the vertically stowed test joint was subject to contact by loads during crane activity; it's position on the perimeter of the catwalk was not protected. The hazard of load contact to the vertically stowed test joint was not identified when deciding its storage location; this was missed during the risk assessment process. The risk assessment process was inadequate.

Additionally, the crane ball was not centred over the load after the joint had been stabilized prior to resuming the lift. The flagger was positioned on the catwalk stairs to stay clear of load path. The deck was congested, requiring flagger to seek less than optimum position to safely perform flagger duties. Ship was rocking due to sea conditions from inclement weather.

Per Job By Design and contractor policy, no taglines were used and safe step back zones identified. Policy change driven to keep personnel out of line of fire. Multiple incidents across GoM where personnel hoisted off deck from tag line entanglement. Tagline management and personnel keeping body position outside of entanglement hazards inconsistent. Human performance: high volume of lifts with ever changing variables, e.g., area configurations, adjacent equipment, etc., mandating constant assessment and awareness to prevent entanglement.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Need to reinforce material safe lifting and maintain safe distance of banksman to avoid coming into line of fire.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

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DATE: 29 Sep 2023
COUNTRY: USA
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
RULE: Safe mechanical lifting

NARRATIVE:

While laying down joint onto pipe wrangler, joint separated from lifting sub, falling approximately 38 ft, striking worker.

WHAT WENT WRONG:

Joint became separated and fell.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

The connections for any BHA components that are new, or which have been broken out and will be re-used/rerun in the well, should be cleaned and visually inspected by WSS for damage and to confirm thread compatibility prior to making up the new BHA to be run in the well.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 25 Jul 2023
COUNTRY: USA
FUNCTION: Drilling
CAUSE: Struck by (not dropped object)
ACTIVITY: Drilling, workover, well operations
RULE: Safe mechanical lifting

NARRATIVE:

The crew just completed setting a tubing plug at 11,485' and were in the process of coming out of the well with wireline. At 400' above the plug (11,085'), the wireline truck had a failure, and the wireline started free spooling into the well. During the free spool, the crew attempted to slow the wireline's descent using the pack off on the lubricator due to the air brakes not functioning. The wireline stopped and gained slack. The crew was now at a stopping point, and while the WSR was seeking guidance on the path forward, the crew decided to clamp the wireline to secure it. While attempting to clamp the wireline, the line became tight, causing the ground sheave to rise up, striking the IP on the right side of the neck/throat area. This resulted in a laceration to that area.

WHAT WENT WRONG:

- Hazard not recognized.
- HSE Procedures or Safe Work Practices did not exist.
- Safeguards that did not work.

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CORRECTIVE ACTIONS AND RECOMMENDATIONS:

During this investigation there was a little bit of mixed information that was discussed. The statements from witnesses had conflicting information. Also, reps from the company were defensive during the investigation as they may have thought there was the potential to get run off. These are just some things to note when doing a severe injury investigation with a small company.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)

DATE: **10 Jul 2023**
COUNTRY: **USA**
FUNCTION: Drilling
CAUSE: Unspecified - Other
ACTIVITY: Drilling, workover, well operations
RULE: Safe mechanical lifting

NARRATIVE:

While tripping in the hole with ¾ inch rods, the rod transfer wire rope cable parted, resulting in the rod transfer bottle falling approximately 85 feet to the ground.

WHAT WENT WRONG:

Approximately 25 feet of wire rope and the attached hook assembly was still connected to the transfer bottle. The rig operator recognized the cable had parted and alerted the floorhand to clear the area. The dislodged rod transfer bottle landed on the ground adjacent to the cellar board on the operator's side of the rig, roughly 5 feet from where the floorhand had been positioned. When the rod bottle landed, it bounced toward the centre line of the rig and came to rest on the cellar board between the wellhead and the operator's station. The operator was approximately 7-8 feet from where the rod bottle landed and remained at the controls. The wire rope coiled around the bottle as it bounced and came to rest near the wellhead. There were no injuries associated with the incident.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Evaluate potential improvements to the rod transfer safety device testing sequence and train on any changes.
- Confirm that the rod transfer safety device start-up and testing procedure is placed in each doghouse near the rig operator.
- The well site supervisor should verify that all crew members have reviewed the rod transfer safety device start-up and testing procedure prior to performing applicable tasks.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 05 Dec 2023
COUNTRY: USA
FUNCTION: Drilling
CAUSE: Unspecified - Other
ACTIVITY: Drilling, workover, well operations
RULE: Energy isolation

NARRATIVE:

Accessing upper level of the pipe skate to retrieve a dropped tubular without fall protection or the skate being isolated.

WHAT WENT WRONG:

No PPE for the task and no proper isolation.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Not available.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: 30 Mar 2023
COUNTRY: USA
FUNCTION: Production
CAUSE: Explosion, fire or burns
ACTIVITY: Maintenance, inspection, testing
RULE: Line of fire

NARRATIVE:

Electrician responded to survey a failure within the H-Pump VFD (Variable Frequency Drive). As electrician was troubleshooting the VFD, worker experienced an ARC flash within the cabinet of the VFD.

WHAT WENT WRONG:

Electrician was troubleshooting the Variable Frequency Drive (VFD) when experienced ARC flash.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Ensure control cabinets are clear of loose conductive material and that finger guards have been installed during PSSR.
- Add questions to PSSR checklist.
- Survey existing facilities to ensure finger guards are installed and cabinets are clear of loose conductive material.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Personal Protective Equipment not used or used improperly

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

NORTH AMERICA ONSHORE

DATE: 31 Oct 2023
COUNTRY: USA
FUNCTION: Production
CAUSE: Exposure electrical
ACTIVITY: Maintenance, inspection, testing
RULE: Energy isolation

NARRATIVE:

Automation technician was troubleshooting a gas lift injection meter building heater. After identifying and isolating the labelled circuit breaker, the technician proceeded to verify zero voltage on the circuit. After verifying zero voltage, the technician inadvertently disturbed a spare set of energized wires causing an arc flash within the panel. No injuries occurred with this near miss.

WHAT WENT WRONG:

Technician discovered the spare wires had been terminated to a mis-labelled breaker and not to the heater block causing the heater end of the unterminated wires to be energized. During this same time, the technician discovered the primary set of wires had been terminated to the heater block but not to the breaker, allowing the primary circuit to not be energized.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Perform spot checks on previous new build locations to verify correct labelling and terminations.
- Add "received and reviewed QA/QC documents from electrical contractor" to PSSR form.
- Enhance RFP and cable testing documents to include 100% point to point verifications.
- Review expectations with electrical contractors during the enhanced QA/QC rollout.
- Order tic tracers to identify energized circuits for employees.
- Continue WMS Appendix - Electrical clarifications to I&E group.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 15 Mar 2023
COUNTRY: USA
FUNCTION: Production
CAUSE: Falls from height
ACTIVITY: Maintenance, inspection, testing
RULE: Working at height

NARRATIVE:

While closing hard-to-reach valve, worker lost footing and fell to landing below.

WHAT WENT WRONG:

Worker lost footing causing them to fall.

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CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Ensure that correct valve handles are accessible and assess what other valve handles should be readily available throughout the field and design and implement designated storage areas near each identified valve.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Personal Protective Equipment not used or used improperly

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: **08 Sep 2023**
COUNTRY: **USA**
FUNCTION: Production
CAUSE: Struck by (not dropped object)
ACTIVITY: Maintenance, inspection, testing
RULE: Energy isolation

NARRATIVE:

While working on the compressor, technician was struck by the valve cap that had released under pressure injuring his hand and wrist.

WHAT WENT WRONG:

Valve released under pressure.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Implement LOTO verification process for 3rd party vendors at central facilities prior to vendors completing work.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

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DATE: **08 Dec 2023**
COUNTRY: **USA**
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Maintenance, inspection, testing
RULE: Unspecified

NARRATIVE:

Tank battery fire.

WHAT WENT WRONG:

Unknown.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Implement interim actions to manage ongoing risk in low flow facilities: visual confirmation on lit flares and flame arrestor heat gun temperature checks.
- Implement engineering actions.

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- Identify low volume facilities for tanks and flares abandonment.
- Expedite the execution of Flare and Tank Removal project.
- Implement engineering changes to prevent standing flames in low pressure flare systems that are not candidates for removal.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

DATE: 18 Aug 2023
COUNTRY: USA
FUNCTION: Construction
CAUSE: Dropped objects
ACTIVITY: Construction, commissioning, decommissioning
RULE: Work authorization

NARRATIVE:

A 100-foot flare stack was erected by a construction crew at a greenfield site. The foreman stopped the job when suspecting guy-wire length(s) required adjustment and developed plan to transition only 2 of 3 guy-wires to temporary concrete anchors (would complete 3rd later that day). Crew completed 2 guy-wires, demobbed crane, and suspended work while proceeding to 2nd work front to set a vessel. However, one worker proceeded (knowingly against instructions and without knowledge of others) to attempt to transition 3rd guy-wire transfer.

WHAT WENT WRONG:

Worker did not recognize crane had been demobilized and was no longer suspending flare. Worker loosened guy-wire clamp causing the flare to fall. The worker was acting without work authorization.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Update Construction foreman handbook to establish:
  - expectation for installation instructions to be utilized on site, and
  - expectation for Work Authorizer to stop work and initiate MOC when encountering or proposing a design/construction methodology change.
- Work Authorizers are to ensure hold points during transitions include communicating that nobody is allowed to stay at or return to previous work front without re-authorization of LSRs.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Use of Protective Methods: Disabled or removed guards, warning systems or safety devices

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

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DATE: 04 Sep 2023
COUNTRY: USA
FUNCTION: Unspecified
CAUSE: Falls from height
ACTIVITY: Maintenance, inspection, testing
RULE: Working at height

NARRATIVE:

While transitioning from the elevated work area to descend an 8ft step ladder, crew member failed to make contact with the ladder and dropped to the deck level.

WHAT WENT WRONG:

Crew member failed to make contact with the ladder causing his fall.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Not available.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 28 May 2023
COUNTRY: Canada
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
RULE: Safe mechanical lifting

NARRATIVE:

Drilling maintenance team were completing planned maintenance on the knuckle boom crane to remove the luffing cylinder pins with a hydraulic jack and pin puller arrangement. The team attempted to remove pin with a 30-ton hydraulic jack once, and a 60-ton hydraulic jack three times without success. On the 5th attempt with a 100-ton hydraulic jack, a 3 ft 15 lbs portion of the threaded rod ejected from the pin puller arrangement, traveling 19 horizontal metres and 35 vertical metres from the crane lifting pin.

WHAT WENT WRONG:

- Threaded rod failed due to exposure to force greater than yield strength - both 60-ton and 100-ton hydraulic jacks were capable of exerting this level of force.
- Crew was not aware of yield load rating or recommended operating limits for the hydraulic jacks.
- Procedures did not identify hydraulic jack to be used, load rating of the provided rod, or potential hazard of threaded rod parting.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Revise procedures/job aids to include specifics on load rating and safe operating pressures specific to threaded road and hydraulic jack being used and recommended actions if safe pin removal efforts are unsuccessful.
- Ensure pin-pulling equipment is independently verified as fit for use prior to mobilizing.
- Consider alternative equipment designs to reduce risk of threaded rod failure.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

NORTH AMERICA OFFSHORE

DATE: **07 Oct 2023**
COUNTRY: **Mexico**
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well operations
RULE: Line of fire

NARRATIVE:

While finishing the riser drain process and preparing to lift them from HCP wellhead deck to rig floor, a shackle fell from CTU to the HCP Mezzanine deck. At that time, personnel were in the CTU area to verify the passage of the risers. No injuries to personnel, nor environment or facility damage.

WHAT WENT WRONG:

Unsafe acts/sudden technical failures.

- Personnel were carrying out simultaneous activities that conflicted with the activity carried out in the mezzanine area.
- The shackle was not lifted with the proper tool.
- The personnel in the CTU and mezzanine were not notified of the shackle lifting, leaving the personnel exposed in the line of fire.
- The worker does not inspect or visualize the risk of the rope breaks getting down the shackle to the lower level.

Context of work/failure.

- There is no supervisor for the activities being carried out in the Moon Pool and CTU area.
- The task of securing the kill line hose should have been planned before removing the tray to begin the Riser lifting operations.
- The personnel working in the CTU do not have radios to maintain communication with the personnel working in the mezzanine area or with the drilling floor.
- The distribution of radios at each level was not planned before the activity.

Organizational factors.

- Procedure removal of spacer reels does not contemplate the sequence of securing the hose and removing the tray as previous tasks.

Barriers/Defences.

- SWA policy is not applied when performing simops.
- The ecological trays are not placed in the CTU area to avoid falling objects at different levels.
- The ATS only contemplates the lifting activity of removing Risers, and does not specify the activity of getting down tools or equipment by lifting between levels, nor does it contemplate the activity of securing the kill line.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Corrective actions:

- Prepare ATS for each level in which tasks are developed and being authorized sequentially to avoid simultaneous tasks.
- Add to the procedure 'removal of spacer spools' the following steps:

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- Check people out of the fire line:
  - Order and cleanliness ensuring isolation with tray in CTU.
  - Securing tools and equipment.
  - Carry out DROPS inspection (Check list).
  - Assign a supervisor with a portable radio to each level.
  - Have set bucket devices to transfer hand tools at different levels.
  - Sequence of securing the hose and removing the tray as previous tasks.
- Add for future contracts a set bucket with retainers for the movement of hand tools (shackles, studs, etc.).

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 19 May 2023
COUNTRY: Mexico
FUNCTION: Production
CAUSE: Explosion, fire or burns
ACTIVITY: Maintenance, inspection, testing
RULE: Energy isolation

NARRATIVE:

During spool welding activities upon arrival of tank from the closed drain, ignition occurred inside the tank, causing the cover to detach. No injuries to personnel.

WHAT WENT WRONG:

Unsafe acts/sudden technical failures:

- The welding activity began on the spool linked to the tank, which contained flammable gases.

Context of work/failure:

- There was no positive isolation when welding started. (Personnel removed a blind flange that was at the beginning of the tank process to present and weld the spool).
- The CCA does not include all the necessary isolation to perform a safely weld. (Blind flange at the tank entrance).
- The CCA does not contemplate a blind flange at the tank entrance and an open vent.
- Tank was not inerted or washed (previously, a leak test with nitrogen and cleaning with Vactor equipment was carried out).

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- The personnel believed that the equipment was inerted and washed.
- Supervisor assigned to the weekend, designed as specialist authority for this activity, does not have the skills to supervise the activity.

Organizational factors:

- Technical and safety skills and competencies required were not identified as critical by the organization to assign resources to shift, weekends, and out of hours activities.
- The process of delivering equipment to the contractors doesn't define the formal communication of its status (washed/inerted).

Barriers/Defences:

- According to interviews, the personnel involved in the activity did not perceive the risk of welding with the tank linked to the spool.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Provide energy isolation training in accordance with procedure where the effectiveness of the training is validated for personnel who perform the functions in the work permit system.
- Update the Area Authority approval/revalidation process for each of the Work Permit System Certificates.
- Define a document that establishes roles and responsibilities of personnel on shift and resources assignment according to task criticality (ILA).
- Incorporate into the training process the technical and security skills validation, according to a competencies by role and function matrix (LTO), including revalidation frequency.
- In the energy isolation procedure, define the equipment delivery format (equipment handover) to be implemented, indicating its conditions (type of cleaning performed, nitrogen sweeping, inerting, etc.)
- Implement the inclusion of a delivery instruction at the planning meeting for confined space/hot work with open flame.
- Acquisition of equipment/lock out tag out accessories.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

NORTH AMERICA OFFSHORE

DATE: 18 Jun 2023
COUNTRY: Mexico
FUNCTION: Unspecified
CAUSE: Unspecified - Other
ACTIVITY: Transport - Water, incl. marine activity
RULE: Unspecified

NARRATIVE:

When carrying out barite transfer operations with an offshore supply ship, on the starboard side with the drilling platform, in DP mode, it was observed that the DP-OS2 (slave) presents a voltage and frequency anomaly. This resulted in a total blackout and the Automatic DP and Manual control were disabled. This caused the vessel to impact on the starboard side with leg #2 of the drilling platform, the mast of the ship's bow with the cantilever of a ladder on the production platform HCP, and the starboard bow with the defence of the same structure.

WHAT WENT WRONG:

Unsafe acts/sudden technical failures:

- When blackout occurred, control of the vessel was lost in DP system and Manual mode system (the emergency generator did not work).
- At the blackout moment, the captain was not on the bridge.

Context of work/failure:

- Lack of leadership on the bridge (Captain). Creating confusion on the instructions for the emergency response (Role of Steering Loss), causing loss of time in emergency care.
- About the first officer, at the time of incident, it was his second day on board and also in the company.

Organizational factors:

- Steering Loss Procedure (blackout) does not specify roles nor instructions to proceed safely.
- On the previous drill's results, there were no findings or any observation or pending actions (follow Hokchi format).

Barriers/Defences:

- According to the interviews and the evidence in the report, it was concluded that the auxiliary generator was not available when the blackout occurred.
- Generator inspection Report was not shown at the time. The report presented did not have the name of the technician neither his signature. The technician was contacted by phone without success due to a medical treatment.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Steering loss Procedure was updated (blackout). Roles and responsibilities were defined and includes the step by step in case of loss position in any scenario.
- Perform steering loss emergency drill by blackout every month with all scenarios, assign roles and responsibilities, and ensure the critical equipment functioning (Hokchi emergency drill evaluation guide and continuous improvement).
- Update and perform the Drill Guide for vessel liaison with JU. Including instructions as: all personnel must arrive to the muster point, waiting time, damage assessment, evacuation case, call-out role.

NORTH AMERICA OFFSHORE

- All personnel on duty at bridge must have Full DP certification. Incorporate in the SACC matrix.
- Verify and implement the correct operation of the ship's maintenance management system. It must be approved by CLASE. Include in contracts.
- Ratify with the contractors that all critical personnel must be approved by MO Hokchi.
- Implement periodic inspections by an accredited third party. Contractual management (initially quarterly).
- Periodic visits to the vessels as part of the work control.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Use of Protective Methods: Disabled or removed guards, warning systems or safety devices

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 12 Aug 2023
COUNTRY: USA
FUNCTION: Drilling
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Drilling, workover, well operations
RULE: Line of fire

NARRATIVE:

Rig operations at the time of the event involved tripping pipe in the hole on the main side drill floor. The injured party (IP) was drifting pipe from a platform located above the MPT carousel. Issues with the racker developed and the worker moved to an adjacent platform for a better visual of the racker. The driller, unaware that the IP changed positions, proceeded to lower the top drive. The top drive trolly assembly struck the IP and caught him between the top drive roller assembly and the platform handrail.

WHAT WENT WRONG:

Visibility of the equipment and personnel was limited by the configuration of the equipment. The driller was unaware of the location change and the location of the platform in proximity to the top drive allowed the IP to be in the line of fire when the equipment was in motion.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Communication - The driller was unaware of the IP's movement to alternate platforms. The platform where the event occurred is only used for maintenance so access to this area is atypical and previously required a work permit. The IP was unaware of requirements to access. The proximity of the equipment to the handrail of the platform created a Human Machine Interface risk that was unprotected.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

NORTH AMERICA OFFSHORE

DATE: **24 Jan 2023**
COUNTRY: **USA**
FUNCTION: Drilling
CAUSE: Falls from height
ACTIVITY: Lifting, crane, rigging, deck operations
RULE: Line of fire

NARRATIVE:

During a routine crane lift, the Crane Operator began to hoist the load per Banksman's direction via radio as the riggers moved into the designated safe stepback area (underneath the pipe skate). One of the Roustabouts inadvertently stepped into the coiled portion of the tagline. Once realizing that a leg was entangled, the Roustabout grabbed and held onto the tagline as the load was being lifted. The banksman observed the event, stopped the job, and the worker was lowered to the deck without injury.

WHAT WENT WRONG:

Excess tagline was coiled and laying on the deck near the load during the lift and worker remained in close proximity as the load was hoisted.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

High percentages of Short Service Employees can result in less experienced workers who are at different places in the training process. Excess tagline can entangle employees, even when tangle-resistant tag lines are used.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

DATE: **08 Jul 2023**
COUNTRY: **USA**
FUNCTION: Production
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Lifting, crane, rigging, deck operations
RULE: Line of fire

NARRATIVE:

The Fabric Maintenance crew was working off a spider basket underneath the Lower Deck to perform sand blasting operations. While relocating the spider basket (located on the West Side Crossmember), the injured person was attempting to run the support pipe through the spider basket choker wire loop located just above the grating. During this operation, the basket operator reported he thought he heard the call to take up slack on the line. The basket operator then shifted tension from the primary wire to the support pipe and as the line went under tension, the finger of the contractor was subsequently pinched in the loop of the choker wire (between the deck grating and wire). The contractor's right middle finger suffered an evulsion from around the first knuckle out to the fingertip.

WHAT WENT WRONG:

- The IP and an assisting coworker both reported that no commands had been given to the basket operator.
- Service Company Spider Basket procedure does not cover communication between the spider basket operator and rigger.

NORTH AMERICA OFFSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Conduct desktop audits of offshore worksite paperwork at a frequency to be determined, but not less than two (2) audits per quarter, per installation.
- Incorporating a comprehensive risk assessment process directly into our safety procedures. Service company Spider Basket procedure to cover proper communication among crew.
- Develop a process to identify effective communication among employees working at a job site to ensure safety, coordination, and seamless operations.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

RUSSIA & CENTRAL ASIA ONSHORE

No high potential events reported.

RUSSIA & CENTRAL ASIA OFFSHORE

No high potential events reported.

SOUTH & CENTRAL AMERICA ONSHORE

DATE: 18 Jan 2023
COUNTRY: Argentina
FUNCTION: Drilling
CAUSE: Falls from height
ACTIVITY: Drilling, workover, well operations
RULE: Working at height

NARRATIVE:

The shift was assembling the conductive pipe to drill the guide. Placing the BOP floors in the proper position, the operator observed that the lower end of the floor support did not enter correctly into the fixing socket to the substructure. He climbed onto the BOP floor and the connection at the upper end of the floor support broke, causing the operator to fall towards the warehouse.

WHAT WENT WRONG:

Organization factors:

- Improper Design.

Job/Fault Context:

- No procedure for assembly.
- The structure does not have a Maintenance Plan.

Unsafe acts and/or sudden technical failures:

- Operator climbed to the floor deployed and not secured, without fall protection elements.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Improve design:

- Get the maintenance plan in place.
- Perform a Risk analysis for the specific task.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Use of Protective Methods: Personal Protective Equipment not used or used improperly

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

SOUTH & CENTRAL AMERICA ONSHORE

DATE: 17 Oct 2023
COUNTRY: Argentina
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Production operations
RULE: Unspecified

NARRATIVE:

Level gauge glass collapse in dehydration tower during normal operation, detected by F&G system in plant.

WHAT WENT WRONG:

- LG used improperly to drain fluid to another vessel.
- LG not catalogued when installed.
- Lack of procedures for its maintenance.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Respect vendor's recommendations for LG installation.
- Catalogue LG and develop maintenance procedures for its inspections/maintenance respecting vendor's recommendations.
- Reinforce the use of SOPs and the need of risk analysis when a new operation mode is needed.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 12 Jun 2023
COUNTRY: Argentina
FUNCTION: Construction
CAUSE: Struck by (not dropped object)
ACTIVITY: Transport - Land
RULE: Driving

NARRATIVE:

Rollover of contractor vehicle on an internal oilfield road.

WHAT WENT WRONG:

- The driver performed an inadequate manoeuvre when he tried to enter a well location (inadequate speed).
- The vehicle checklist shows degradations that were not repaired.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Verify contractor's vehicle maintenance programme.
- Verify contractor conductor's performance and defensive driving course compliance.

SOUTH & CENTRAL AMERICA ONSHORE

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: **30 Jun 2023**
COUNTRY: **Brazil**
FUNCTION: Production
CAUSE: Struck by (not dropped object)
ACTIVITY: Maintenance, inspection, testing
RULE: Energy isolation

NARRATIVE:

After installing a drain line extension in a pig chamber located at São Roque Station, the chamber was pressurized to check its tightness and a small leak was identified from the pig chamber lid pin. During depressurization of the chamber, carried out by its new drain, to allow the repair of the leak from the cover pin, the drain line whipped, which hit the worker who was positioned diametrically opposite the open end of the drain and approximately 4 metres away from the drain valve. The impact of the line hit the worker's helmet and the movement of the helmet's temple caused a minor injury to his scalp.

WHAT WENT WRONG:

- Norms, standards, and good engineering practices were not considered in planning, construction, installation, and decommissioning: need to move the drain discharge without definitive information on the location or project for adaptation.
- Change management procedure with inadequate results: change management carried out for the flow reversal process did not include the suitability of the pig launcher for the pig receiver, nor was it completed with the necessary document updating or with the updating of information in related systems.
- Inadequate planning: in the "Work authorization" questionnaire, there was a mistaken negative marking about it being a change, leading the team to make a mistake.
- Identification and qualitative or quantitative analysis of risks non-existent: Level 1 Hazard Analysis was not carried out.
- Inadequate identification of steps: ineffective coverage of a similar accident that occurred in another installation, not generating a review of the standard that allows depressurization of gas pipeline pig chambers through the drain.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Lessons learned:

- Non-routine activities with construction and assembly characteristics, with any change, must be preceded by a Change and Project Management process.
- The scope of accident actions must be analysed and disseminated by the Multidisciplinary Commission for the Coverage of Safety, Environmental, and Health Anomalies of the Production Unit.

SOUTH & CENTRAL AMERICA ONSHORE

Recommendations:

- Define and formalize with the contracted company the installation location of the containment dike for the drain of the pig receiving chamber and its waste sampling point for analysis; implement a project to adapt the pig receiving chamber, in accordance with Standard; formalize and disseminate guidance for maintenance teams to only perform "construction and assembly" activities through Change and Project Management.
- Carry out the review to adapt the documentation of the duct system and pig chambers (flowcharts, drawings, updating data in computerized systems).
- Retrain supervisors and engineers in the "Work authorization" and "Manage changes in industrial facilities" standards, with a focus on identifying changes to systems and processes.
- Review the item related to pig launch and receipt of the "gas pipeline operation" standard, in line with what is established in the Standard, and train technicians and operators in this new version of the standard.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

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DATE: 27 Jul 2023
COUNTRY: Ecuador
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Transport - Land
RULE: Driving

NARRATIVE:

A truck was reversing, loaded with stony material to deposit it on the ground as part of the maintenance of the access road to a telecommunications tower. The ground gave way, causing erosion under the tires and subsequent overturning of the truck. The driver was not injured. The truck suffered damage.

WHAT WENT WRONG:

- The truck was reversing with its tires on the edge of the road, without identifying the tire track or the possibility of soil erosion.
- Poor identification of hazards and risk assessment, it does not identify the hazard associated with soil stability and risk of overturning.
- Lack of supervision at the work front and the driver does not have the required competencies.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Cultural Management depends on the commitment of all levels, to maintain a high sense of vulnerability, through operational discipline.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

SOUTH & CENTRAL AMERICA ONSHORE

DATE: **01 Jul 2023**
COUNTRY: **Peru**
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Unspecified - other
RULE: Unspecified

NARRATIVE:

During the maintenance handover operation of a valve, the manual shut-off was closed, resulting in a gas leak from one of the vents. The evacuation alarm was triggered, and the line was depressurized, leading to the shutdown of the plants.

WHAT WENT WRONG:

Maintenance management, inspection, reliability: Incorrect inspection process or cycle. The execution of none of its stages (pre-commissioning, commissioning, PSSR).

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

During plant shutdowns, make sure to correctly execute the inspection and testing of ball valves according to established procedures.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: **01 Apr 2023**
COUNTRY: **Peru**
FUNCTION: Unspecified
CAUSE: Unspecified - Other
ACTIVITY: Unspecified - other
RULE: Unspecified

NARRATIVE:

Worker, while performing maintenance, noticed smoke coming from the adjacent area in the camp kitchen and subsequently saw fire on the roof. They proceeded to use a nearby extinguisher and immediately evacuated the area.

WHAT WENT WRONG:

Inadequate Installation: The gas supply line was found to have a loss of hermeticity.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Inspect and perform thorough maintenance of supply lines and odorizer for operability.

SOUTH & CENTRAL AMERICA ONSHORE

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 02 Mar 2023
COUNTRY: Trinidad & Tobago
FUNCTION: Production
CAUSE: Falls from height
ACTIVITY: Maintenance, inspection, testing
RULE: Other issue – no applicable rule

NARRATIVE:

A work party of five persons were engaged in conducting replacement of PSVs on the V200 vessels. The valves are located on a steel grated platform approximately 31ft above ground level. Access to this platform is via a vertical fixed ladder. Due to the known corroded steel grating, the flooring was previously overlaid with scaffold planks. During mobilization of tools, two technicians were on the platform one of which was pulling up tools while standing on the scaffold plank. The scaffold plank under him broke and the grating was unable to support the contractor. This resulted in him partially falling through the grating and coming to a stop around his upper torso area. The contractor extricated himself and both technicians descended the platform.

WHAT WENT WRONG:

- The team, in preparing the risk assessment for the PSV replacement activity, did not identify the defective plank atop the PSV rack as an impaired mitigation.
- The scaffold planks placed on the platform to mitigate against the corroded grating risks were ineffective at the time of the incident. The scaffold planks were installed over 5 years ago and since then they were not monitored.
- The extent of the corrosion of the grating on the PSV rack was not known because the defects on this rack were not identified by the various integrity surveys conducted in previous years.
- The defective grating was not replaced in a timely manner. An initial maintenance order was created for the grating anomaly but was cancelled without being properly revalidated and reassigned.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Develop a self-verification programme to verify that the site readiness meeting meets its intent and objectives of site being ready and resourced each shift to safely deliver the upcoming activity set.
- Increase the AA area authority resources on the site to manage the activity set and associated CoW requirements.
- Develop a mitigation management process that includes monitoring and replacement of mitigation controls for structural anomalies.
- Conduct integrity surveys at site of all areas previously inaccessible to identify extent of corrosion and corrective action where necessary.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

SOUTH & CENTRAL AMERICA OFFSHORE

DATE: 04 Aug 2023
COUNTRY: Brazil
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well operations
RULE: Other issue – no applicable rule

NARRATIVE:

While racking back a stand of 6 5/8" Drill pipe on Aux Well centre the stand was intended to be racked in the last slot of Row 43. When fingers were functioned to be open, the latch on the upper fingerboard failed to open correctly. When racking the stand back, the pipe came in contact with the closed latch and sheared it from its cylinder and its secondary retention wire. This resulted in a dropped object with the potential for a fatality. The latch weighs 2.4kg and fell from a height of 33m.

WHAT WENT WRONG:

- inadequate/defective warning systems/safety devices
- inadequate hazard identification or risk assessment
- inadequate supervision

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Implement improvements for the real-time monitoring systems.
- Update the monitoring system management procedure.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 25 Jul 2023
COUNTRY: Brazil
FUNCTION: Production
CAUSE: Struck by (not dropped object)
ACTIVITY: Maintenance, inspection, testing
RULE: Energy isolation

NARRATIVE:

During boiler work on the pipe of the electrostatic treater relief system for the installation of flanges, blockers were inserted downstream and upstream of this segment of the pipe to isolate the presence of hydrocarbons. During the piping preparation activity for flange installation, the blocker projected out of the line, hitting the boilermaker in the hip region, causing a fracture.

WHAT WENT WRONG:

- Inadequate communication between work groups: from the identification of the need to the integration of the activity into the executive shutdown schedule, the information did not flow completely, meaning that the activity was not detailed to the level of mentioning the use of blockers.

SOUTH & CENTRAL AMERICA OFFSHORE

- Inadequate selection criteria: no hiring criteria were identified regarding the number of qualified professionals for the installation and operation of blockers.
- Non-existent change management: base maintenance planning was structured with the participation of security technicians preparing the land Level 2 Hazard Analysis. As a result, many Level 2 Hazard Analysis are carried out without the participation of those requesting the service and without the participation of production operators.
- Ambiguous or confusing instructions and requirements: in the interpretation of the people involved in the work permit in which the accident occurred, different companies can work together on the same work permit, even without a bond or contractual relationship between them.
- Errors are not detectable: during the accident analysis process, the committee identified that the Work authorization description was changed after Level 2 Hazard Analysis was validated by the safety technician and after going through the simultaneity meeting.
- Failure to identify the change: it was identified that the oil operator position was being performed by an operator who was not trained and there was no workforce change management. It was also seen that the acting supervisor on the shift in which the accident occurred did not have workforce change management.
- Inadequate level of detail: the contracted company's standard for the use of blockers does not have clear information regarding the positioning of the blocker vent hose and is not emphatic regarding the participation of the executing team in releasing the work permit, so that all risks related to this activity are discussed on site.
- Inadequate refresher training: it was not possible to demonstrate from the training certificate that the contracted company emphasizes the need for the participation of the specialist technician in releasing the work authorization.
- Preventive or corrective actions for insufficient or inadequate performance: some non-conformities were identified in the work authorization audit process.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Lessons learned:

- All work authorizations must have all risks discussed with all performers at the time of release of the service.
- It is essential to manage change in the workforce.
- Activities that require the use of special tools must be included in the details of the scheduled shutdown executive schedule.
- The Safety Manual checklists need to be used and known by everyone involved in the work authorization process.

Recommendations:

- Evaluate the possibility of creating a functional scheduled shutdown committee to identify good practices, including everything from the list of services planned to be carried out during a scheduled shutdown, planning details, integration of executive schedule tasks, to carrying out the activities during the days of scheduled maintenance shutdown.
- Suggest to the supply area the creation of hiring criteria related to the number of qualified professionals for the installation and operation of blockers, which are compatible with the activity planning.
- Carry out a critical analysis of the production unit's base maintenance planning system, identifying possible changes in the application of onboard procedures and the best way to execute them on land.
- Suggest to the manager of the Work authorization standard to clarify the requirement related to the execution of work composed of employees from two contracted companies without any link or contractual relationship between them.

SOUTH & CENTRAL AMERICA OFFSHORE

- The Work authorization management system must determine the return to the Level 2 Hazard Analysis validation stage by the security technician whenever there is a change in the description of the service to be performed at the Work authorization.
- The Production Unit's Functional Change Management Committee must carry out a critical analysis of why failures are occurring in identifying the change in workforce.
- Suggest to the company contracted for the use of blockers to carry out a critical analysis of its installation procedure and safe operation of isolation and line testing tools.
- Emphasize in training the mandatory participation of the specialist technician in the release of work permits.
- Systematically evaluate Work authorization audit reports within the scope of the Production Unit, checking deviations that recur throughout the periods, propose recommendations to address them, and control the effective implementation of the recommendations.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: 20 May 2023
COUNTRY: Brazil
FUNCTION: Unspecified
CAUSE: Unspecified - Other
ACTIVITY: Transport - Water, incl. marine activity
RULE: Other issue – no applicable rule

NARRATIVE:

A platform supply vessel was carrying out a diesel supply operation for a drillship, when the Dynamic Positioning Officer (DPO) on duty noticed that the vessel showed a loss of heading of 5 degrees, moving the stern away from the Unit.

Diesel pumping was immediately interrupted through the emergency stop and the DPO took manual control, stabilizing the vessel so that the hose could be disconnected and returned to the drill ship.

There was no loss of integrity of the onboard hose and manifold and there was no diesel oil leak into the sea or onto the vessel's deck.

WHAT WENT WRONG:

The checklist does not include specific steps to check the risks of interference caused by the thrusters of the Maritime unit.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Review the Dynamic Positioning (DP) Manual to include instructions on the configuration of thruster allocation for an operation scenario with marine units in Dynamic Positioning with the possibility of being within the thruster water discharge zone.

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- Review the Configuration, Approach and Departure Checklist in DP to adapt the instructions on checking the risks of interference caused by thrusters during operations with maritime units operating in DP.
- Issue a safety alert with lessons learned, covering other vessels under contract.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 25 Jul 2023
COUNTRY: Trinidad & Tobago
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Production operations
RULE: Line of fire

NARRATIVE:

At approximately 04:45, the bonnet holding bolts for a 2-inch globe valve, operating at 960psi, failed, resulting in a gas release. An emergency shutdown was initiated, and the platform was safely de-pressured.

Once the gas had dissipated, the investigation team found that the actuator and other components had been dislodged, causing damage to the piping and cooling fans above. The actuator was found ~5ft away from valve body location with other components up to 25ft away. Cause of valve failure was CSCC (chloride stress corrosion cracking) of SS304 bolts.

WHAT WENT WRONG:

- Project documentation did not prohibit the use of SS304 materials – they were "not recommended".
- Project Quality controls did not prevent delivery and installation of a valve with SS304 bolts that were stated in supplier documentation to be SS316.
- Operations integrity management documentation did not provide guidance for inspection of stainless-steel materials.
- Inspection and replacement campaigns resulting from past SS304 failure events had a narrow focus and so failed to identify full scope of the risk.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Update Project standards to require that stainless steel fasteners must have ≥ 2% Mo content.
- Update the procedure to provide guidance on stainless steel inspection and also include inspection of all stainless-steel bolting in the inspection contractor's scope.
- Conduct gas detector coverage modelling to confirm existing gas detectors provide sufficient coverage.
- All assets: review SS304 risk and create action plan to manage this.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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For the full analysis of 2023 results, as well as all fatal incident reports submitted to IOGP since 1991, visit IOGP's data website at: https://data.iogp.org

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